Book of Abstracts

7th International Symposium on

Plant Molecular Farming

Programme

ISPMF 2026

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Universitat Politecnica de Valencia | Spain

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Madeinplant

UNIVERSITAT POLITÈCNICA DE VALÈNCIA

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About this Book of Abstracts

This book contains all 111 accepted contributions to ISPMF 2026, grouped by presentation type and ordered by presentation code. Day and time of each presentation are indicated. Flash talks (PF) appear once in their own section and are not duplicated in the poster section.

Section	Codes	Count
Keynote Lectures	K-01 - K-08	8
President's Prize Talk	**PP-01**	**1**
Oral Presentations	O-01 - O-28	28
Flash Talks	PF-01 - PF-15	15
Poster Presentations	P-01 - P-59	59
TOTAL		**111**

KEYNOTE LECTURES

K-01 Wednesday 13 May | 10:00-10:45

Plant Molecular Farming - a 360° review

Sutinee Soopairin*, Mathew Paul, Suthira Taychakhoonavudh* and Julian Ma****

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Topic: Plant-Made Biologics

Plant molecular farming (PMF) has been a promising manufacturing platform for over three decades, but has yet to make a significant commercial breakthrough. The closure of large and important PMF companies in recent years was a major blow to confidence and interest in plant manufacturing technologies, leaving PMF in a potentially existential crisis. The purpose of this study was to assess attitudes towards PMF around the world, from within and without the field and to identify common themes to inform a forward strategy for PMF. To this end, we have conducted two interview studies. The first was conducted using detailed semi-structured interviews with 63 people involved in recombinant protein production, including senior academics, early career researchers, CEO/CSOs from PMF companies, representatives from the mainstream pharmaceutical industry, government, major funders and private investors. In order to include the public voice, the second study involved shorter semi-structured interviews with 48 members of the public. Data was analyzed using framework analysis to identify perceived advantages of PMF, major concerns, and regional differences in the way forward to PMF success. This talk will discuss the key findings, highlighting areas where there are fundamental differences in opinions between different stakeholders, such as the key advantages of plant manufacturing platforms, selection of target products and potential contribution to global health. The talk will also address issues around general awareness and understanding of PMF and future recruitment of young scientists to the field. The future of PMF will require not only addressing barriers but also strengthening collaboration, improving communication, and demonstrating credible success stories to build global interest and confidence toward PMF. The ISPMF community is invited to participate in this ongoing work, providing mutual support to develop a strategy for the future of PMF, to formulate a strategic action plan, and become involved in implementing that plan.

K-02 Wednesday 13 May | 15:20-16:05

NovoBodies: Next-generation AI-Designed Antibody Mimetics

Alfredo Quijano

Monod Bio

Topic: Molecular Farming Marketplace

At Monod Bio, we've developed the NovoBody platform, which leverages AI-enabled computational protein design to generate novel binding molecules that exploit the binding interfaces of existing antibodies or antibody fragments. The process eliminates suboptimal properties of the original molecules while enhancing key functionalities without requiring extensive discovery campaigns. The result is small, single-chain, highly stable molecules that are easy to manufacture in both prokaryotic and eukaryotic systems, with customizable properties and formats. We have applied this design approach to generate multi-specific and high-valency constructs with a unique chain (4-8 binding sites) for diverse life science applications.

K-03 Thursday 14 May | 09:00-09:45

From Plants to Planets: Plant Virus Therapeutics for One Health

Nicole F. Steinmetz

UC San Diego

Topic: Viral Vectors & Nanoparticles

We turned toward plant viruses as a platform nanotechnology - re-engineered to solve challenges in human health, agriculture, and even space exploration; an approach toward One Health. Because plant viruses can be produced efficiently through plant molecular farming using plants as bioreactors, they offer a sustainable, scalable platform for next-generation biotechnology. We have developed plant virus-based delivery systems packaging small molecules or biologics (proteins and nucleic acid therapeutics). Given their immunomodulatory nature, we have developed vaccine and cancer immunotherapy candidates targeting cancer and other chronic diseases. A lead candidate for intratumoral immunotherapy has undergone testing in canine cancer trials treating companion dogs and is now entering the clinical development pipeline. Another avenue is the repurposing of plant viruses to enable plant health; we employ principles of nanomedicine to target plant pathogens with foliar and soil applications. Finally, we extend plant virus biotechnology beyond Earth. Plant molecular farming in space could enable astronauts to grow and produce medicines on demand during long-duration missions. In this lecture, I will discuss the engineering design rules governing plant virus-based nanoscale biologics and show how these principles enable translation across biomedical, agricultural, and space-based applications toward a One Health paradigm.

K-04 Thursday 14 May | 11:55-12:40

Molecular farming for 25 years - aligning our barley bioreactor to business opportunities

Bjorn Orvar

ORF Genetics

Topic: Molecular Farming Marketplace

This year ORF Genetics is celebrating its 25th anniversary. ORF Genetics is a plant molecular farming company that has developed a production platform for recombinant proteins using bioengineered barley seeds as its bioreactor. Since launching its first product on the market 18 years ago, the ISOkine growth factors for stem cell research, the company has been continuously improving its gene technology, hydroponic greenhouse cultivation as well as the protein processing technology, to meet the demand for increased yield, higher volume, and lower production cost. The company has also gained valuable experience and continues to test in-field cultivation as an important tool to lower production cost. As plant systems for recombinant protein production arrived late on the scene and are still novel to many industry players and stakeholders, the company has put a strong emphasis on business development, quality assurance, and customer service. With new gene technology and data applicable for plants, such as genome sequencing and DNA editing, the future for molecular farming in plants is bright, especially when it comes to demand for high volumes of recombinant proteins needed for large-scale cell cultures and biomedical application.

K-05 Thursday 14 May | 15:50-16:35

Scientifically validated new cosmetic ingredients from plants and plant cells

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Topic: Metabolic Pathway Engineering

The EU-financed InnCoCells project focused on developing innovative and sustainable plant-based production processes for the exploitation of scientifically validated cosmetic ingredients based on underutilised plant resources. The final aim was to bring at least 10 new cosmetic ingredients to the pre-commercial stage. Four different production platforms were investigated: plant cell cultures, hairy roots, aeroponics, and field-grown plants. We also investigated a cascade biorefinery concept in which by-products and biowaste from the agri-food industry were utilized to extract further bioactive molecules. Overall, 90+ plant species were cultivated in one or more of these systems. We also applied systematic approaches such as metabolic engineering to optimise growth conditions and the yields of target compounds. The optimised processes were demonstrated by pilot-scale production and downstream processing. The ingredients and extracts were evaluated using advanced chemical analysis and a unique panel of innovative enzymatic, cell-based in vitro, and human tissue-based ex vivo assays to validate their efficacy and safety. The production processes were characterised by techno economic analysis and life cycle assessment to ensure economic feasibility and a small environmental footprint. By the end of the project, four ingredients had already been commercialized, and more than 10 ingredients had reached the pre-commercial stage.

K-06 Friday 15 May | 09:00-09:45

Plants as Biofactories of Metabolites: Breeding and Engineering for High-Level Production in Tomato

Antonio Granell

IBMCP (CSIC-UPV)

Topic: Metabolic Pathway Engineering

Plants are inherently efficient producers of specialized metabolites. Many native metabolites that accumulate across plant species have been selected during evolution because they enhance fitness and adaptation to specific environments, often by improving tolerance to biotic and abiotic stresses. At the same time, numerous plant metabolites have valuable applications in food, health, and industry. While natural genetic diversity can be exploited through selection and breeding to increase the production of specific metabolites, the species that naturally synthesize a desired compound may be difficult to cultivate. In such cases, engineering the relevant metabolic pathways into a more agronomically amenable species offers an attractive, sustainable alternative, especially when microbial fermentation is not feasible. In our laboratory, we have used plants--primarily tomato--as biofactories for added-value, health-promoting compounds, while also demonstrating that fine-tuning metabolite production in tomato fruit can enhance both flavor and nutritional quality. In this presentation, I will discuss examples where breeding can be used to increase or decrease the levels of target metabolites affecting the nutritional and organoleptic traits of tomato. I will also present cases where genome editing and metabolic engineering enable the high-level production of non-native or naturally low-abundance metabolites with added value.

K-07 Friday 15 May | 11:35-12:20

Molecular farming activities within the framework of GMO and genome editing regulations

Ana Judith Martin

Spanish Ministry of Agriculture, Fisheries and Food (MAPA)

Topic: Molecular Farming Marketplace

Molecular farming involves the use of biotechnology in the production of valuable proteins, small molecules, and peptides. Organisms that possess a novel combination of genetic material obtained through the use of modern biotechnology are considered living modified organisms according to the definition of Cartagena Protocol on Biosafety. Based on the above, molecular farming activities and products obtained through these activities may be subject to regulations on genetically modified organisms (GMOs). These regulations imply different requirements based on the organism, techniques and containment measures. In addition, specific provisions and requirements are included for food and feed uses containing, consisting of and produced from GMOs. Modern biotechnology has been advancing at a very rapid pace with new molecular techniques. Genome editing technologies are currently amongst the most promising ones and genome edited plants are increasingly being developed globally. Therefore, several countries are developing a policy of genome edited plants based on the Cartagena Protocol, which has involved among other actions, the revision and adaptation of their regulatory framework to keep pace with scientific developments while maintaining a high level of protection of human, animal and plant health and the environment. In conclusion, the regulations applicable to these activities and their products are of great interest to all parties involved in molecular agriculture, from the initial stages of development to commercialization and use.

K-08 Friday 15 May | 14:40-15:25

Toward commercialization of a plant-based VLP seasonal influenza vaccine, and beyond

Marc-Andre D'Aoust

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Topic: Molecular Farming Marketplace

Plant-based platforms have enormous potential as manufacturing technologies, but this potential has long been held back by technological and commercial barriers. Aramis Biotechnologies was founded in 2023 with the objective of breaking these barriers and unlocking the full potential of plant-based bioproduction. As a first step toward this objective, Aramis is currently working on bringing the first plant-based seasonal influenza human vaccine to commercialization. To this aim, we have acquired the virus-like particle (VLP) product from Medicago and successfully optimized it by stabilizing the hemagglutinin antigen in pre-fusion conformation for the 3 seasonal strains and improved the process to increase the density of antigens on VLPs. Preclinical immunogenicity studies have confirmed the increased immunogenicity of the optimized product. The purification process was also improved, increasing the overall process yields by up to 20-fold. The optimized VLP seasonal influenza vaccine has now entered clinical development with a phase 1/2 immunogenicity study being conducted in Canada to compare its immunogenicity to that of commercial influenza vaccines. Beyond influenza, Aramis also seeks to leverage its technological and manufacturing leadership position with strategic partnerships to expand its presence across broad applications of the platform, including pharmaceutical, veterinary and agricultural applications.

PRESIDENT'S PRIZE TALK

PP-01 Wednesday 13 May | 10:45-11:15

Context-dependent Pattern-triggered Immunity Control of Agrobacterium-mediated Transient Expression Efficiency

Yinuo Nora Liu* (1), Nattapong Sanguankiattichai (1), and Renier A. L. van der Hoorn (1)

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Topic: Smart Chassis

One major limitation to the broader application of agroinfiltration is inconsistent protein yield. Previous work suggests that pattern-triggered immunity (PTI) mounted during early infection may limit transient expression efficiency, as depletion of pattern recognition receptors (PRRs) that trigger PTI against Agrobacterium increases protein yield. Moreover, agroinfiltration of recalcitrant plant species often fails to induce cell death associated with late immune responses, but still results in undetectable protein production, suggesting suppression by early non-specific immunity. We therefore tested the impact of PTI on transient expression and Agrobacterium virulence by artificially inducing PTI during agroinfiltration in N. benthamiana. Acute PTI triggered by the flg22 elicitor peptide had no effect on Agrobacterium virulence or protein yield. In contrast, sustained PTI generated by pre-expression of Agrobacterium-recognising PRRs significantly decreased Agrobacterium virulence and protein yield. To explain this difference, we monitored early PTI outputs and found that Agrobacterium suppresses flg22-triggered reactive oxygen species bursts in a concentration-dependent manner. Together, these results show that PTI conditionally inhibits Agrobacterium-mediated transient expression: Agrobacterium actively suppresses acute elicitor-triggered PTI, while sustained immune activation remains a barrier to efficient transient expression.

ORAL PRESENTATIONS

O-01 Wednesday 13 May | 11:45-12:05

A Novel Eco-Friendly Biopolymer from Phaseolin

P. Creanza* (1), M. Osnato (1), E. Maricchiolo (1), S. Scataglini (1), F. De Marchis (2), M. Bellucci (2), A. Aluigi (1), G. Beihammer (3), J. Schwestka (3), E. Kapusi (3), E. Arcalis (3), E. Stoger (3) and A. Pompa (1).

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Topic: Plant-Made Biologics

Petroleum-based materials remain a primary source of global pollution, driving the scientific community to develop sustainable, eco-friendly bioplastics. This study proposes an innovative strategy to partially replace conventional plastics using phaseolin (P), an edible vacuolar protein that constitutes 50% of the protein content in common bean seeds. Naturally, phaseolin forms homotrimers but cannot assemble into extended polymers. To overcome this, we genetically modified its coding sequence by introducing a cysteine codon at the C-terminal tail (P*). This modification enables the formation of intermolecular disulfide bridges, facilitating the assembly of P* units into long polymeric chains. The P*$ gene was introduced into the N. tabacum plastidial genome via biolistic transformation. Plasticization tests of the resulting protein confirmed the formation of resilient, thin biopolymer films. The engineered P* was further modified by inserting additional CYS to produce a biopolymer that could increase polymer size. To compare with the nuclear transformation, a different strategy stategy was tested, such as adding the KDEL peptide to increase the P* retention inside the endoplasmic reticulum (ER) , which facilitates the production of the disulfide bridges thanks to its oxidizing environment. Future applications of this new biopolymer are expected across different sectors, ranging from packaging for the food industry to the biomedical sector as a carrier of protein-based drugs.

O-02 Wednesday 13 May | 12:05-12:25

The use of plant-derived cancer immunotherapics to provide treatments for pregnant women, while avoiding fetal and neonatal harm

YJ Rosenberg*(1), S Permar (2), L Haigwood (3), ME Ackerman (4), K Van Rompay (5).

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Topic: Plant-Made Biologics

Recent PlantVax studies comparing transfer efficiency of plant-derived HIV-specific mAbs and their high (FcRn) affinity YTE and LS-mutants with mammalian counterparts following passive SC infusion into macaque dams close to parturition, demonstrated that while plant- and mammalian-derived Abs exhibited comparable efficiency across the yolk sac in mice, plant-derived Abs were unable to cross macaque placentas. This defect was closely associated with altered Fc glycan expression of plant-derived Abs and poor Fc■R binding and triggering, suggesting a second Fc■R receptor is required for maternal-fetal IgG transfer of across three-layer primate placentas and the use of plant-derived Abs as a means of providing cytotoxic treatments for woman during pregnancy, while avoiding fetal harm. Accordingly, highly successful anti-cancer mAb treatments, Rituximab, Trastuzumab and Cetuximab for B cell lymphomas, breast cancer and rectal cancer respectively, were produced in plants and administered to pregnant macaque dams prior to delivery. As predicted, pharmacokinetics showed high mAb levels in dams with no transfer of mAbs to babies. In addition, B lymphocytes were absent in dams but present in variable but normal levels in newborns. These results indicate a potential role for plant-derived immunotherapeutics in preventing side effects in babies; including known B cell depletion in infants following Rituximab and respiratory complications in babies born of Trastuzumab treated mothers

O-03 Wednesday 13 May | 12:25-12:45

FROM LEAF TO LIFESAVER: Nicotiana benthamiana as a highly versatile platform for glycoprotein-based vaccines against parasitic worms

GM Bakker*, A. Schots, and RHP Wilbers

LABORATORY OF NEMATOLOGY, WAGENINGEN UNIVERSITY & RESEARCH, THE NETHERLANDS

Topic: Plant-Made Biologics

Infections with the parasitic worms Schistosomes and hookworms have tremendous impact on human health. Vaccines could provide a sustainable control strategy but are not available to date. Current vaccine research is mainly hampered by antigen selection and recombinant antigen production with appropriate post-translational modifications, like glycosylation. Recently controlled human infection studies were exploited to select new antigens, which we produced in our Nicotiana benthamiana expression platform. The versatility of this platform allowed us to quickly screen a range of antigens for both parasites. After selecting proteins that were highly expressed and secreted into the apoplast, we focused on their glycan composition. We produced these glycoproteins with defined N-glycans found on secreted parasite proteins from different life-stages. To do this, we (1) developed a CRISPR workflow that allows us to efficiently knock-out undesired plant glycosyltransferases or glycosidases and (2) co-expressed exogeneous glycan-modifying enzymes with our vaccine antigens. Using our platform we were able to produce Schistosoma mansoni Cathepsin B1, with a variety of N-glycans. These plant-produced Cathepsin B1 variants are used to study how different N-glycans contribute to antigen recognition and protective immunity and are even investigated in a vaccination challenge model. These studies showcase how plants can fuel research towards novel anti-parasite vaccines.

O-04 Wednesday 13 May | 12:45-13:05

Plants-on-a-chip: microfluidics-based real-time tracking of protein body formation in plant cells

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Topic: Plant-Made Biologics

Zein protein bodies are specialised storage organelles naturally occurring in maize seeds. Due to their high chemical resistance and immunostimulatory properties, they represent promising vehicles for oral vaccine delivery. This application is further promoted by their distinct multi-layered structure, which enables the in vivo bio-encapsulation of therapeutic proteins. Despite the successful ectopic formation of multi-layered protein bodies in leaf tissue of Nicotiana benthamiana , the mechanisms underlying their assembly remain poorly understood. Current models based on static immunolocalization studies suggest a sequential assembly of different zeins in this process, however a time-resolved analysis is still missing. Here, we report in situ protein body biogenesis in real-time. In our approach Nicotiana tabacum BY-2 cells are utilised for transient transformation with fluorescently labelled zein genes via the plant cell pack platform. The cells are subsequently loaded into a microfluidic chip for continuous cultivation. This setup allows the in vivo monitoring via fluorescence microscopy, providing insights into the spatiotemporal dynamics of protein body assembly. This unprecedented view can be important for the assessment and expansion of existing models on protein body formation. The technology has further application potential to assess, for example, the growth of plant cells lines in different media conditions and cultivation optimization.

O-05 Wednesday 13 May | 13:05-13:25

A plant-produced FcR-agonistic lectibody targeting high-mannose glycans for cancer immunotherapy

KL Mayer (1) and N Matoba (1, 2, 3)

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Topic: Plant-Made Biologics

Glycans are critical regulators of disease pathogenesis and immune function, yet they remain largely underexploited as therapeutic targets. N-linked high-mannose (HM) glycans are aberrantly enriched on the surface of malignant cells and enveloped viruses, where they contribute to disease progression and immune evasion. To target this disease-associated glycobiomarker, we developed a "lectibody" composed of the HM-specific lectin Avaren fused to the human IgG1 Fc region (AvFc), produced using a Nicotiana benthamiana transient expression system. We previously showed that AvFc selectively recognizes clustered HM glycans on virus-infected cells and cancer cells, leading to Fcγ receptor engagement and activation of cytotoxic effector functions. In this study, we evaluated the antitumor activity of AvFc in a syngeneic orthotopic ovarian cancer model using the murine ID8 cell line in C57BL/6 mice. AvFc treatment (25 mg/kg, every other day for 15 doses) significantly prolonged median survival. Cytometry by time-of-flight (CyTOF) analysis revealed expansion of total splenic leukocytes and increased CD8+ T cells in the peritoneal fluid, suggesting that AvFc's antitumor activity is partially mediated by adaptive immune responses. Together, these findings demonstrate the therapeutic potential of a plant-produced lectibody and highlight molecular farming as a viable platform for generating novel Immunotherapeutics.

O-06 Wednesday 13 May | 14:40-15:00

Using in planta glycoengineering to decode mucosal structure-function relationships of human secretory IgA

Jacqueline Seigner, Gudrun Kucher, Richard Strasser, Kathrin Goritzer

Department of Biotechnology and Food Science, BOKU University, Vienna

Topic: Plant-Made Biologics

Secretory IgA (SIgA) is a promising modality for mucosal therapeutics owing to its potent pathogen neutralization, anti-inflammatory activity, and stability in secretions. A defining feature of SIgA is its extensive N- and O-glycosylation, yet tissue specific glycan profiles and their functional significance remain incompletely understood. We addressed this gap by profiling site-specific glycosylation on human SIgA from saliva, colostrum, and milk, and used transient expression and glycoengineering in Nicotiana benthamiana to assemble SIgA with tailored, homogeneous glycans. Our toolkit enabled controlled presentation of human relevant motifs, including highly sialylated variants characteristic of colostrum and milk, and Lewis-type arm-fucosylation found in saliva, features not readily accessible in conventional mammalian systems. We further installed defined mucin-type O-glycosylation in plant lines with reduced endogenous O-glycan backgrounds. Across biochemical, biophysical, and cell-based assays, hinge O-glycosylation primarily modulated protease resistance with minimal impact on receptor engagement, while specific N-glycan features shaped commensal binding and stability in saliva and simulated gastric fluid. The ability of established plant platforms to produce high quality SIgA with defined glycosylation provides unprecedented insight into SIgA structure-function relationships and advances its therapeutic potential.

O-07 Wednesday 13 May | 15:00-15:20

Development of a transgene-free edible vaccine model via mobile mRNA fusion

DS Kayihan*(1, 3), C Kayihan (2, 3), G Lomonossoff (3)

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Topic: Plant-Made Biologics

Traditional vaccines require qualified healthcare personnel for administration and expensive equipment due to purification procedures, transportation and storage, making access to these vaccines difficult for developing countries. Edible vaccines, besides sharing the advantages of plant-based recombinant vaccines, have further benefits like lack of purification steps, easier administration and increased stability. However, producing edible vaccines via transgenic methods brings limitations such as the risk of transgene escape and low public acceptance. Along the plant vascular system, mRNAs with specific mobility motifs can be transported from source to sink tissues where they are translated. It is well established that some of these mobile mRNAs are also able to pass through the graft junctions. This study aims to combine hetero-grafting technique with mobile mRNA fusion to develop a transgene-free edible vaccine model. For this purpose, three mobile mRNA constructs were generated by fusing GFP with mobility motifs of tRNAmet-deltaDT, Flowering Locus T and gibberellic acid-insensitive RNA and infiltrated to Nicotiana benthamiana leaves by means of Agrobacterium to investigate their capability of transport and translation in non-infiltrated leaves. The most efficient motif will be fused to SARS-COV-2 RBD and transiently expressed in N. benthamiana grafted with sweet-basil and tomato scions potentially capable of producing antigens free from transgene and Agrobacterium.

O-08 Thursday 14 May | 09:45-10:05

Delivering RNA to mammalian cells using a plant virus capsid

GP Lomonossoff* (1), H. Peyret(1,2), SN Shah (1), Y. Meshcheriakova (1), L. Mitchenall (1), H. Martin (3), DL Tomlinson (3)

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Topic: Viral Vectors & Nanoparticles

RNA molecules of defined sequence have many potential uses in biomedicine and bionanotechnology. However, one of the main factors limiting their deployment is their intrinsic instability, meaning that preparations must be stored at low temperature. As a consequence, several approaches have been investigated to stabilise the molecules, including the incorporation of non-natural nucleotides into the sequence and encapsulation of the molecules.; The plant virus, cowpea mosaic virus (CPMV), has icosahedral particles that are extremely effective at protecting its genomic RNA. As a result of our studies on the virus replication cycle, we demonstrated that RNA replication and encapsidation are tightly coupled. We have utilised this to develop a method for the encapsidation of designer RNAs into CPMV particles and have shown that the encapsidated RNA is stable for months at room temperature. CPMV particles harbouring RNA encoding GFP can be taken up by mammalian cells and the RNA released and translated. This uptake can be greatly improved through the bispecific affimer reagents which can bind the virus particles and cell-surface proteins. Furthermore, preliminary studies in mice suggest that the administration of CPMV particles containing RNA encoding the Zikavirus structural proteins can elicit the production of anti-Zika antibodies. These studies suggest that CPMV particles could be an effective approach to delivering mRNA molecules to cells without the need for cold storage.

O-09 Thursday 14 May | 10:05-10:25

Effective production of protein nanoparticles in two plant-based systems

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Topic: Viral Vectors & Nanoparticles

Protein nanoparticles self-assemble into large quaternary structures and are innovative tools in biotechnology, however, further work is required to develop them in plant-based systems. We have investigated the plant-production of four protein nanoparticles as scaffolds for antigen display, each with properties affecting their suitability for certain applications. To expand our toolbox of plant-made nanoparticles we investigated ferritin from Helicobacter pylori, which assembles from 24 identical subunits. Ferritin accumulated to over 0.2 mg/g of fresh weight in Nicotiana benthamiana, and genetic fusion of a viral glyco-epitope to this protein unexpectedly tripled the accumulation. The fusion protein was efficiently N-glycosylated and assembled into nanoparticles. Chimeric and wild type ferritin showed similar iron mineralization, which was reduced by introducing substitutions at two critical amino acids. This adds ferritin to our suite of plant-made nanoparticles for vaccine development. To allow for high throughput screening of multiple combinations of nanoparticles and pathogenic viral and bacterial immunogens, we adopted the plant cell pack (PCP) technology, using BY-2 cell aggregates in microtiter plates. Three protein nanoparticles were produced in PCPs with accumulation up to 0.38 mg/g. Scale-up for purification and characterization of these proteins was successful in 1000-fold larger PCPs, demonstrating the potential of the PCP screening and production platform.

O-10 Thursday 14 May | 10:25-10:45

Immunoengineering in space: pioneering VLP-based immunotherapies produced through plant molecular farming in low resource environments

Patrick Opdensteinen (1), and Nicole F. Steinmetz (1)

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Topic: Viral Vectors & Nanoparticles

Plant virus-like particles (VLPs) have remarkable immunomodulatory properties, allowing these biologics to be repurposed as precise, safe, and adaptable vaccines and immunotherapies against cancer and chronic diseases. Their modular, self-assembling nature enables the rapid production of diverse formulations from minimal genetic templates - further, the fact that VLPs can be produced using simple infrastructure and plant molecular farming, makes these VLPs ideal candidates for on-demand biomanufacturing in low-resource or extreme environments. In this work we focused on addressing the manufacturing challenge, specifically focusing on availability and purification for on demand use in low resource environments, i.e. space flight. This fusion of biotechnology and space research exemplifies the next frontier of "off-world biopharmaceuticals", supporting human space travel to mars and beyond.

O-11 Thursday 14 May | 10:45-11:05

Nanobody-functionalized potato virus X nanoparticles for targeted cancer therapy and protein delivery

E. Lozano-Sanchez* (1), M.A. Moreno-Gonzalez (2), F. Merwaiss, (1), N.F. Steinmetz (2), and J.A. Daros (1).

(1) Instituto de Biologia Molecular y Celular de Plantas (Consejo Superior de Investigaciones Cientificas - Universitat Politecnica de Valencia), Valencia, Spain. (2) Department of NanoEngineering, University of California San Diego, La Jolla, California, USA.

Topic: Viral Vectors & Nanoparticles

Plant virus nanoparticles (VNPs) is a promising material in nanomedicine thanks to its inherent biological and physicochemical properties. VNPs can be manipulated through different methods to display molecules of interest for biotechnology applications. In previous studies, we have set up a genetically encoded potato virus X (PVX) system for displaying nanobodies on the VNP surface for SARS-CoV-2 neutralization. Here, we further explore the application of this system for cancer targeted therapy and protein delivery. First, we developed different PVX-derived VNPs decorated with different nanobodies against HER2, one of the most common cancer-related markers. The combination of nanobody targeting abilities with the intrinsic antitumor properties of PVX nanoparticles supports the potential of this VNP in cancer therapeutic applications. The targeting capacity of the nanoparticles were validated in vivo in solid tumors. Second, we leveraged the binding capacity of nanobodies displayed in VNPs to create a protein delivery system. We present a proof of concept consisting of nanobody-decorated VNPs able to bind the receptor-binding domain (RBD) of the SARS-CoV-2 Spike protein. Once loaded with the protein of interest in vitro, they can be used as a delivery system with potential applications in vaccine development. In summary, our results reaffirm the versatility and potential of VNPs for nanomedicine applications, particularly in targeted cancer therapy and protein delivery.

O-12 Thursday 14 May | 11:05-11:25

Plant-Derived VLPs for Cancer Immunotherapy: Coupling Neoantigen Packaging with APC-Targeted Display

A Utard, A Guigand, M Demeules, C Ott, , L Bellott, A Trolet, V Poignavent*

Serendip innovations SAS

Topic: Viral Vectors & Nanoparticles

Virus-like particles (VLPs) have achieved clinical success as prophylactic vaccines against hepatitis B, HPV and influenza. Their nanoscale size (<100 nm) and highly ordered, repetitive protein structure enhance interactions with B cells and antigen-presenting cells (APCs) compared with soluble antigens, enabling potent immune activation with an established safety profile. Despite these advantages, the ability of VLPs to induce cytotoxic CD8+ T-cell responses for cancer immunotherapy has been limitedly evaluated clinically, outside settings such as HPV-driven malignancies. Here, we present a plant-derived VLP platform based on Grapevine fanleaf virus (GFLV) that enables simultaneous engineering of both the inner and outer particle surfaces. We designed GFLV VLPs to encapsidate tumor-derived neoantigens within the inner cavity while displaying APC-targeting moieties on the outer surface to enhance uptake and antigen presentation. In murine preclinical models, the engineered VLPs demonstrated APC-targeting capability and therapeutic vaccination activity, supporting induction of tumor-directed cellular immunity. Overall, this plant-VLP platform provides a versatile scaffold for next-generation therapeutic cancer vaccines, combining modular payload/display design with the manufacturing advantages of plant bioproduction and the potential to drive robust CD8+ T-cell responses.

O-13 Thursday 14 May | 12:40-13:00

Pilot-scale production facility of clinical-grade therapeutic nanoparticles

A Rosa*(1), A Doglia (1), L Rinarelli (1), Roberta Zampieri (1), Linda Avesani (2)

Diamante SB srl (1), Universita degli Studi di Verona (2)

Topic: Scaling Up Plant-Based Production

Plant-based biomanufacturing of therapeutic proteins represents a relatively novel platform which provides several advantages compared to other systems, including cost-effective scalability, simplified upstream processes and inherent biosafety. Diamante SB has developed a pilot-scale GMP-compliant manufacturing process where Nicotiana benthamiana serves as a bioreactor to produce therapeutic nanoparticles derived from Tomato Bushy Stunt Virus (TBSV) and engineered to display a non-native immunodominant peptide necessary to induce immune tolerance in patients affected by Rheumatoid Arthritis. Manufacturing consists of two main phases: the upstream process entails three consecutive cycles of infection with replication-competent viral particles, in a vertical farming layout that increases productivity per unit area. Downstream purification, on the other hand, has been simplified to consist of simple and labour-friendly centrifugation steps to obtain a high-purity active ingredient. At present, the facility produces more than 15Kg of biomass per year, with an average yield of 71mg per Kg of leaf fresh weight and higher-than 97% purity. While manufacturing has been streamlined to obtain fast and consistent production quality, certain hurdles remain to optimize scalability with minimal losses. In this sense, Diamante SB is in the development stage of an optimized process with the goal to increase yield linearity and overall quality of its therapeutic ingredient.

O-14 Thursday 14 May | 13:00-13:20

Recombinant Protein Expression Reached 30 g/kg in Oryza Sativa endosperm

Kunpeng Li, Ningli Wang, Heng Ying, Yong Yang, Ming Luo, Daichang Yang and Cliff Yang

Healthgen Biotech. Inc., Wuhan, Hubei, China

Topic: Scaling Up Plant-Based Production

Human blood has remained the only source for human serum albumin since the 1940s. OsrHSA, a recombinant human serum albumin purified from bioengineered Oryza Sativa endosperm, was approved by China National Medicinal Products Agency (NMPA) in 2025 as the first rice-derived pharmaceutical in the world. Although OsrHSA possesses much higher purity (99.9999%) than blood-derived albumin (~96%), it faces a commercial challenge of a very low market price (~$5 per gram). Here we report a series of genetic engineering techniques which increased the recombinant albumin expression to ~30g/kg of brown rice, which is 37.5% of all host proteins contained in rice. First, we discovered a natural endosperm-specific Gt13a promoter. Second, we utilized a low glutelin content rice transgenic line. Third, we further enhanced the Gt13a promoter by removing its negative regulatory motifs. This new expression technology enabled OsrHSA to compete commercially with blood products.

O-15 Thursday 14 May | 14:30-14:50

Unlocking unique products whilst maintaining industry standards

A Schaaf*

Topic: Scaling Up Plant-Based Production

Rising structural and mechanistic complexity in next-generation biologics--driven by deeper disease-pathway insights and the need for defined post-translational modifications such as glycosylation--often exceeds the capabilities of conventional microbial and mammalian hosts. These systems struggle particularly with proteins requiring extensive disulfide bonding, multidomain organization, or multiple glycosylation sites, creating manufacturing bottlenecks. Plant-based expression systems provide an alternative route for such molecules. Moss is presented as one example that supports correct folding, human-like glycosylation, and expression of complex proteins. This includes recombinant Factor H (CPV-104)--a ~150-kDa complement regulator comprising 20 SCR domains, ~40 disulfide bonds, and eight N-glycosylation sites--which is in clinical development, underscoring that plant-derived production can meet the quality requirements for advancement to human studies. From a process standpoint, a 500-L production scale is well established and has been scaled out, with further scale-up activities underway, , indicating a clear path to larger-scale, GMP-compatible manufacturing. Overall, plant-based technologies may help address key constraints in producing advanced biologics by enabling the manufacture--and clinical progression--of proteins that are otherwise difficult to express in traditional systems, as exemplified by the clinical-stage CPV-104 program.

O-16 Thursday 14 May | 14:50-15:10

Reinventing Sweetness: How Plant Molecular Farming Unlocked the Industrial Production of Protein Sweetener Thaumatin

C. Jimenez (1,2) & Y. Gleba (1,2)

Nambawan Spain SLU (1), Nomad Bioscience GmbH (2)

Topic: Molecular Farming Marketplace

Excessive consumption of sugar imposes a severe global burden on human health -- contributing to overweight, obesity, diabetes, cardiovascular diseases, and related disorders -- and carries a substantial environmental footprint, requiring more than 33 million hectares worldwide for the cultivation of sugar cane, sugar beet, and sugar corn.; There is a powerful alternative in nature: thaumatin, a naturally occurring sweet protein with an extraordinary sweetness intensity--over 13,000 times that of sucrose on a weight basis. However, the widespread use of thaumatin has historically been limited by the lack of an economically viable industrial production system.; In 2025, Nambawan Spain, a technology spin-off of Nomad Bioscience, successfully brought Thaumatin II to market under the trademark Thauma(TM), using plant biofactories to enable scalable, cost-efficient manufacturing.; Thauma(TM) combines exceptional sweetness and taste attributes with functional versatility: it is non-caloric, non-glycemic, and has received regulatory approval from JECFA, FDA, and FEMA for use as a sweetener and flavor modifier with clean-label status.; In this talk, we will outline the key technical developments that enabled the industrial production of Thauma(TM) and discuss the scientific, regulatory, and market challenges encountered in bringing this novel ingredient to global markets.;

O-17 Thursday 14 May | 15:10-15:30

Scale-up and DoE-based optimization of plant cell packs enables transformation of 600 g BY-2 cells with target yields of more than 1 g/kg biomass

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Topic: Scaling Up Plant-Based Production

The plant cell pack (PCP) transient expression platform enables efficient microplate-based high-throughput screening of recombinant biopharmaceuticals in Nicotiana tabacum BY-2 cell suspensions by removal of the cultivation medium and subsequent infiltration with Agrobacterium tumefaciens. However, once a subset of promising product candidates has been identified, the method fails to provide sufficient protein quantities for detailed functional characterization and pre-clinical studies. Therefore, we scaled the PCPs by 1,000-fold from 60 mg to 60 g using a centrifugation-based approach, yielding ~400 mg/kg fresh mass (FM) of the model protein DsRed, ~300 mg/kg FM of the malaria vaccine candidate antigen PfAMA1 and ~160 mg/kg FM of the human PfAMA1 specific monoclonal antibody 1E4. A further 1,000-fold increase to 600 g was achieved by the technical refinement of vacuum-filtration strategy and devices. Using design of experiments (DoE) we optimized the following process parameters: i) sucrose concentration in the infiltration buffer, ii) co-incubation time of BY-2 and A. tumefaciens prior to PCP packing, iii) PCP thickness as well as iv) residual moisture content of the PCPs and increased product accumulation to >1 g/kg FM DsRed, ~550 mg/kg PfAMA1 and ~280 mg/kg mAb 1E4. Hence, the optimized PCP procedure is suitable to provide sufficient amounts of target protein (>100 mg) for detailed functional characterization and might be further applicable for small-scale production.

O-18 Thursday 14 May | 15:30-15:50

Scale-up and cGMP Manufacturing of Plant-Made Pharmaceuticals

H Haydon* (1), B Bratcher (1), K Whelan (1)

KBio, Inc. (1)

Topic: Scaling Up Plant-Based Production

As one of the few organizations in the world capable of large scale, cGMP manufacture of pharmaceuticals from plants, this presentation will share recent changes in KBio's business structure as well as major advances in production capacity. Using actual product development case studies, KBio's experience in both production scale up and design of quality systems necessary to meet regulatory requirements will be addressed. Included will be a discussion of the unique aspects of implementing cGMP controls when using a whole plant system, quality assurance progression from research, pre-clinical, Phase 1, Phase 2 and Phase 3, as well as cost implications. Experience from direct interactions with both US FDA and European Medicines Agency will be shared to highlight the different approaches to early-stage requirements for clinical manufacturing. The presentation will conclude with practical tips for preparing product candidates for scale up and efficient tech transfer to a CDMO.

O-19 Thursday 14 May | 17:00-17:20

Benthi.x: A gene-edited, space-efficient Nicotiana benthamiana platform for heterologous protein production in closed environments

K LeBreux, B Giroux, L Feyzeau, MC Goulet, C Goulet, and D Michaud*

Centre de recherche et d'innovation sur les vegetaux, Universite Laval, Quebec City, Canada

Topic: Smart Chassis

Closed environments offer practical advantages for the cultivation of plant protein biofactories including plant uniformity, product consistency, water/nutrient recycling and production cycles on a year-round basis. Here, we present CRISPR/Cas9-edited lines of transient expression host Nicotiana benthamiana presenting a compact, space-efficient phenotype well suited to such environments. Our strategy consisted of altering apical dominance by suppressing the synthesis of strigolactone, a negative regulator of bud outgrowth-promoting cytokinins. Strigolactone-depleted lines were produced by hindering the expression of either Carotenoid cleavage dioxygenase 7 (CCD7) or CCD8, two key enzymes of the strigolactone synthetic pathway. Knocking out either enzyme had no impact on biomass production but strongly influenced the auxin/cytokinin ratio, primary metabolism-associated proteome and aboveground architecture. The DeltaCCD (or ‘benthi.x’) lines showed altered metabolic fluxes driving energy and cytokinin production, an axillary growth-oriented pattern and a spatial footprint reduced by 45-50% compared to the LAB strain commonly used for protein expression. Recombinant protein yields were maintained in the benthi.x lines, as here shown with GFP and therapeutic chimeric antibody rituximab. Our data confirm the usefulness of knocking out CCD7 and CCD8 in N. benthamiana to generate space-efficient lines well suited to closed production environments, including vertical farming systems.

O-20 Thursday 14 May | 17:20-17:40

Bioluminescent sentinel plants enable autonomous diagnostics of viral infections

Camilo Calvache (1), Marta Rodriguez-Rodriguez (1), Victor Vazquez-Vilriales (1), Elena Garcia-Perez (1), Aubin Fleiss (2), Mustafa Ezzeddin-Ayoub (3), Fabio Pasin (4), Jose Antonio Daros (1), Karen S. Sarkisyan (2), Diego Orzaez (1), Marta Vazquez-Vilar* (1)

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Topic: Smart Chassis

Plants engineered with synthetic genetic programs can transform how we monitor and manage the extension of crop pests and diseases. Here, we establish a bioluminescent platform in Nicotiana benthamiana for autonomous viral sensing based on the fungal bioluminescence pathway (FBP). We first demonstrate that recombinant viruses can deliver missing pathway components, enabling spatially resolved tracking of infection dynamics. Leveraging this starting point, we developed a dualoutput sentinel circuit that uses a protease-responsive Bioluminescence Resonance Energy Transfer (BRET) module to report infection through a virus-triggered spectral shift in luminescence. In the absence of infection, plants emit a stable yellow glow indicating system integrity. Upon infection with potyviruses, cleavage of the BRET fusion by the virus-encoded NIa-Pro protease activates a distinct colour change detectable with low-cost imaging. This modular design is compatible with other pathogens carrying specific proteases and supports future multiplexing strategies. Our results highlight the potential of synthetic sentinel gene circuits as autonomous biosensors for precision crop protection.

O-21 Thursday 14 May | 17:40-18:00

From Leaves to Seeds: Transcriptional Rewiring and Time-Resolved Monitoring of Recombinant Antibody Accumulation

R Vitek*(1,2), N Šindelarova (1,3), A Hlavata (1,2), T Moravec (1)

1) Laboratory of Virology, Centre for Plant Virus Research, Institute of Experimental Botany of the Czech Academy of Sciences, Prague. 2) Department of Genetics and Breeding, Faculty of Agrobiology, Food and Natural Resources, Czech University of Life Sciences, Prague. 3) Faculty of Food and Biochemical Technology, University of Chemistry and Technology, Prague.

Topic: Smart Chassis

Nicotiana benthamiana is widely used for rapid transient production of recombinant biologics, yet leaf cells are poorly suited for high-capacity protein synthesis and storage, limiting yields. Linking fundamental plant developmental biology with applied biomanufacturing, we test whether leaf cells can be transcriptionally reprogrammed toward a seed-like, storage-competent state to enhance antibody accumulation.; We engineered hybrid transcription factors derived from regulators of embryogenesis, seed maturation, and regeneration, fused to strong synthetic activation domains to boost transcriptional output. Their transient co-expression with antibodies aims to increase biosynthetic activity and remodel lytic vacuoles into protein storage-like organelles.; Protein dynamics are monitored using fluorescent timer proteins that change emission over time, distinguishing newly synthesized from older pools. Combined high-throughput plate assays and confocal microscopy reveal altered compartmentalization and spatial separation of synthesis and storage sites.; While clear cellular remodeling is observed, quantitative gains in accumulation and stability will be validated by biochemical and mass spectrometry analyses. This approach integrates mechanistic insight with practical optimization of plant molecular farming.

O-22 Thursday 14 May | 18:00-18:20

Immune Attenuation of Nicotiana benthamiana against Agrobacterium tumefaciens for Molecular Farming

Emma C. Watts* (1), Kaijie Zheng (1), Evan E. Ellison (2), Renier A. L. Van der Hoorn (1)

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Topic: Smart Chassis

Plant biofactories present an opportunity for sustainable, scalable production of recombinant proteins but currently face a bottleneck in production capacity. In Nicotiana benthamiana, one of the most widely used hosts, yield is limited by the plant's immune response to Agrobacterium tumefaciens, which carries pathogen-associated elicitors even when disarmed. One promising strategy to overcome this bottleneck is to modulate the immune profile of N. benthamiana to minimize immune elicitation. In this work, we have conducted a large-scale knock-down screen of immunity-related genes to identify targets for immune attenuation. In parallel, we are developing stable transgenic lines carrying knock-outs of key immunity genes. The transgenic lines, generated using virus-induced gene editing (VIGE), provide a versatile platform that can be stacked to create an optimized immune profile. We have already seen increased recombinant protein production by up to 5x in knock-down plants related to pattern recognition receptors, salicylic acid production, ethylene production, and potential antibacterial agents. We have generated and validated the associated prp27- and npr1- mutants, where we saw increases of about 2x. We are now generating further mutants and mutant stacks that avoid Agrobacterium recognition to maximize recombinant protein yield, with the aim of making these available to the community.

O-23 Friday 15 May | 09:45-10:05

Integrating elicitation of cell cultures with genome assembly, full-length transcript sequencing and epigenetic modulation to dissect boldine biosynthesis in Peumus boldus

Matus JT*(1), Aviles-Kruuse N (2), Sanchez G (2), Navarro-Paya D (1), Bru R (3), Conessa A (1); Bombarely A (4), Moreno-Romero J (5), Hasbun R (2)

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Topic: Metabolic Pathway Engineering

Peumus boldus is a medicinal tree species native to Chile and the main natural source of boldine, an aporphine alkaloid with well-documented antioxidant, anti-inflammatory and neuroprotective properties. However, boldine production currently relies on leaf extraction from native forests, limiting its scalability and sustainability.; Here, we present an integrative genomic and functional framework to elucidate the regulation of boldine biosynthesis in P. boldus. We generated a first draft genome assembly complemented with Iso-Seq transcriptomes from leaves, flowers, fruits and boldine-producing in vitro cell cultures. This multi-tissue long-read transcriptomic dataset enabled refinement of gene models and identification of candidate genes associated with boldine metabolism. In parallel, we established long-term cell suspension cultures and evaluated the effects of elicitors and epigenetic modifiers on boldine accumulation. Our results show that modulation of chromatin state and stress-related signaling pathways significantly alters boldine production, supporting a regulatory role of epigenetic mechanisms in specialized metabolite biosynthesis. By integrating genome assembly, Iso-Seq-based transcriptomics and functional perturbation assays, this study provides insights into the genetic and epigenetic control of boldine biosynthesis and highlights P. boldus cell cultures as a platform for studying and engineering specialized metabolism in woody plant species.

O-24 Friday 15 May | 10:05-10:25

Recombinant plant-based production of cyclic irregular monoterpene pheromones via a novel trans-alpha-necrodyl diphosphate synthase

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Topic: Metabolic Pathway Engineering

Irregular monoterpenes are high-value C10 compounds widely used in sustainable pest control as key components of insect pheromone blends. Their broader deployment is currently limited by the small number of known irregular isopentenyl diphosphate synthases (IDSs) and by the lack of efficient biobased production platforms. Improving the understanding of irregular IDS enzymatic activity while enabling plant recombinant expression is thus essential for scalable bioproduction of pheromones or their precursors. Within the Pheroplus project, we described a novel cyclic irregular IDS from Lavandula that catalyzes the formation of trans-alpha-necrodyl diphosphate. Its ester derivative, trans-alpha-necrodyl acetate (TNDAc), is a component of the pheromone blend of Delottococcus aberiae, an invasive mealybug pest. The enzyme, named TNDS, was functionally validated both in vitro and in planta. Notably, we developed tobacco plant biofactories producing TNDAc at levels sufficient to attract D. aberiae males under controlled conditions, demonstrating the feasibility of plant-based pheromone production and emission. Comparative analyses with related irregular IDSs point to specific structural features likely involved in determining the synthesis of cyclic versus branched products. These observations support ongoing efforts toward rational enzyme design and optimization, and highlight the potential of combining enzymatic insight with plant molecular farming for sustainable pheromone bioproduction.

O-25 Friday 15 May | 10:25-10:45

Ortholog Driven CRISPRa Modules Enable Enhanced and Flexible Plant Gene Activation

Amir Akhgari*, Aaron M. Klepper, Heiko Rischer

VTT Finland

Topic: Metabolic Pathway Engineering

Programmable transcriptional activation with CRISPRa offers a potent approach to gene regulation for synthetic biology and plant biotechnology. Here, we introduce a modular CRISPRa/Cas ortholog framework built on the multi-kingdom Golden Gate platform that enables direct comparison of dSaCas9 and dSpCas9 for activating genes in Nicotiana benthamiana. Across assays, dSaCas9 consistently produced stronger activation of target promoters than dSpCas9, achieving up to a 19-fold increase in transient dual-luciferase experiments. Beyond the constitutive pNOS promoter, we also assessed several plant-specific promoters with the CRISPRa system; in multiple cases, dSaCas9 again showed superior activation compared to dSpCas9. To expand the system's utility, we adapted it for stable expression in diverse plant cell cultures. We transformed the model N. tabacum BY2 line using standard procedures and further tailored transformation protocols for coffee cell cultures and tomato callus. Cells from all three species were successfully transformed with a construct driving activation of the constitutive pNOS promoter.; Overall, our modular CRISPRa/Cas system supports efficient transcriptional activation both transiently in plants and stably in cell cultures. Ongoing efforts with cell culture platforms focus on accelerating the development of customizable gene circuits and metabolic pathways for plant synthetic biology.;

O-26 Friday 15 May | 10:45-11:05

Compartmentalized metabolic engineering for carotenoid production across biological scales

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Topic: Metabolic Pathway Engineering

Carotenoids are essential lipophilic isoprenoids synthesized in plastids and fundamental for human nutrition and photosynthetic physiology. In green tissues, their tight co-regulation with chlorophylls sustains efficient light harvesting and photoprotection. Yet boosting carotenoids inside chloroplasts can disrupt this balance, sometimes driving chloroplast-to-chromoplast conversion, which is excellent for storage, but detrimental for photosynthetic efficiency. To bypass this trade-off, we investigated strategies to enhance carotenoid biosynthesis outside plastids, following the rationale that compartmentalizing metabolites can prevent harmful intracellular interactions. By engineering the cytosolic mevalonate (MVA) pathway in Nicotiana benthamiana leaves and introducing bacterial enzymes for carotenoid production, we successfully redirected the synthesis of intermediates such as phytoene and lycopene to the cytosol, demonstrating that pathway relocation is feasible. Looking ahead, extraplastidial carotenoid biosynthesis opens new avenues in microalgae, where cellular simplicity and plasticity may further improve production. We focus on the diatom Phaeodactylum tricornutum, combining rapid growth, mature genetic tools, and a highly compartmentalized metabolism shaped by secondary endosymbiosis. Altogether, we hope this work pushes carotenoid fortification beyond the limits imposed by the photosynthetic chloroplast.

O-27 Friday 15 May | 14:00-14:20

Scalable expression technologies for manufacturing bioengineered proteins in plants

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Topic: Scaling Up Plant-Based Production

Transient expression technologies such as magnICON(R) and NOMADIC(R) are efficient processes for biopharmaceutical and biomaterial production that are simple and indefinitely scalable protocols for heterologous protein expression in plants. They rely on transient amplification of viral vectors delivered systemically by Agrobacterium or by viral particles or released from a transgene using ethanol induction. These technologies effectively address most of the major shortcomings of earlier plant-based technologies such as low speed of manufacturing, low expression levels, lack of versatility, and high manufacturing cost. The technologies have been brought to GMP-compliant level and have been successfully used to manufacture diagnostic antibodies or produce materials for clinical trials by Icon Genetics/Bayer, Icon Genetics/Denka, Nomad Bioscience, KBio etc. In addition, Nambawan Spain, together with Nomad Bioscience and Olivos de Badajoz, has successfully scaled up and commercialized the technology based on stable transformation of plants expressing a protein of interest in endosperm of seed. Examples of technologies used at different (lab, pilot and industrial) manufacturing scales will be presented.

O-28 Friday 15 May | 14:20-14:40

Determinants of Public Hesitancy Toward Medicines or Biomolecules Derived from Plant Molecular Farming: preliminary results in a Thai population

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Topic: Molecular Farming Marketplace

Our prior in-depth interviews with 63 stakeholders highlighted concerns regarding public hesitancy toward plant molecular farming (PMF). However, empirical evidence on public hesitancy toward PMF-derived products remains limited. Building on these insights, this study examined public hesitancy toward medicines or biomolecules derived from PMF and associated factors among the Thai population. A cross-sectional survey was conducted in January 2026 using a structured, self-administered online questionnaire based on the Theory of Planned Behavior and distributed via social media to Thai individuals aged 18 years or older. Descriptive statistics were used to characterize hesitancy and regression analysis identified associated factors. A total of 227 participants completed the questionnaire. The median age was 31 years, with respondents from all regions of Thailand. Overall, 54.2% reported hesitancy toward using PMF-derived products, with most showing moderate or higher hesitancy. A positive attitude toward the tobacco industry was associated with a twofold increase in non-hesitancy toward PMF, while a positive attitude toward PMF was also significantly associated with non-hesitancy. These findings indicate that public hesitancy toward PMF-derived products is evident in Thailand and influenced by multiple factors. This pilot study provides a basis for expansion at ISPMF 2026 to engage wider international audiences and inform broader communication and branding strategies for PMF.

FLASH TALKS

PF-01 Wednesday 13 May | 17:30-17:35

Rapid Production of Broadly Neutralizing Antibody CR9114 in Nicotiana benthamiana for Potent Neutralization of H5N1 and H7N9 Influenza Strains.

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Topic: Plant-Made Biologics

Highly pathogenic avian influenza viruses such as H5N1 and H7N9 continue to pose a serious pandemic threat, emphasizing the need for rapidly deployable and broadly protective strategies. This study shows that Nicotiana benthamiana can serve as a rapid-response platform for producing CR9114, a broadly neutralizing antibody targeting the conserved influenza hemagglutinin stem. Plant-produced CR9114 potently neutralized recently isolated 2023 H5N1 strains in vitro, with IC■■ values as low as 0.646 ng/mL, comparable to mammalian-cell-derived antibodies. In contrast, reduced but detectable activity was observed against H7N9, consistent with glycan-mediated steric shielding in Group 2 viruses. Despite limited in vitro neutralization, previous studies demonstrate that CR9114 confers strong in vivo protection through Fc-mediated effector functions. Plant-produced CR9114 displayed a predominantly high-mannose glycosylation profile, which may enhance Fc-dependent immune activity. Overall, this work supports plant molecular pharming as a practical route toward a rapidly deployable intranasal antibody-based intervention for emerging zoonotic influenza threats.

PF-02 Wednesday 13 May | 17:35-17:40

Hergreen as NF-engager: Bispecific antibody for enhanced cancer immunotherapy

Yes

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Topic: Plant-Made Biologics

More than three decades have passed since Hiatt et al. first reported the production of IgG in plants. Although many researchers have attempted to develop monoclonal antibody (mAb) biologics in various plant species, none have yet received FDA or USDA approval for market release. Limitations of plant-based biologics have been identified not only in quantitative terms but also in qualitative aspects. In the case of trastuzumab, we found that near-complete glycoengineering of plant-specific N-glycosylation enhanced the efficacy of the mAb and even resulted in novel biodistribution patterns in a xenografted mouse model. Furthermore, when Hergreen(R) was combined with the in-situ vaccine CPMV virus-like particle (VLP), the survival rate of a syngeneic mouse model reached 100%, suggesting the potential of PhytoRice(R)-expressed mAbs. Using this platform, we further glycoengineered mAbs with human 1,4-galactosyltransferase, as well as other mAbs such as immune checkpoint inhibitors and Hergreenpet. Based on the promising effects of PhytoRice(R), we proceeded to develop Hergreen as NF-engager, Bispecific antibody, and an antibody-drug conjugate (ADC). Hergreen-ADC is being compared its efficacy with Kadcyla and Enhertu. The results of these studies will be discussed at the meeting.

PF-03 Wednesday 13 May | 17:40-17:45

Safety and pharmacokinetic study of a plant-produced Nipah virus neutralising porcine monoclonal antibody

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Topic: Plant-Made Biologics

Nipah virus (NiV) is a highly infectious zoonotic RNA virus that infects pigs and causes acute encephalitis in humans. Passive immunization with neutralising monoclonal antibodies targeting viral envelope glycoproteins has been shown to be an effective therapeutic strategy. A novel neutralising IgG1 antibody (mAb A2), specific for the attachment glycoprotein G (gpG), was isolated from NiV-vaccinated pigs and represents a promising therapeutic candidate. mAb A2 was transiently expressed in ΔXF Nicotiana benthamiana plants by agroinfiltration and purified by protein A-based magnetic beads affinity chromatography, yielding approximately 50 mg/kg fresh weight. Pseudovirus-based neutralisation assays showed that the plant-produced antibody displayed neutralising activity comparable to that of a HEK293T-produced counterpart. Surface plasmon resonance analysis revealed a high gpG-binding affinity in the picomolar range. In addition, the plant-derived antibody exhibited enhanced binding to porcine FcγRIIIa compared to the mammalian-produced version. Pharmacokinetics and biosafety were evaluated in pigs following single intravenous administration at 1 mg/kg and 2.5 mg/kg. Circulating antibody levels remained stable until day 14 and declined linearly by day 28, and no adverse effects were observed. Neutralisation assays on sera are ongoing. Overall, these results highlight the potential of plant-derived mAb A2 as a safe and effective therapeutic candidate against NiV.

PF-04 Wednesday 13 May | 17:45-17:50

Plant-based expression and engineering of Fc-silent bispecific immune checkpoint inhibitors for affordable cancer immunotherapy

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Topic: Plant-Made Biologics

Immune checkpoint inhibitors (ICIs) have transformed cancer treatment but remain inaccessible to many patients in low- and middle-income countries. Transient expression in Nicotiana benthamiana offers a rapid, flexible and easily scalable alternative to mammalian systems while enabling antibody engineering at the levels of glycosylation, Fc function, and molecular architecture.; Here, Fc-engineered monoclonal and bispecific ICIs were produced in wild-type and glycoengineered plants. IgG1 Fc variants containing silencing mutations demonstrated improved accumulation, assembly, and thermostability compared to IgG4P scaffolds. Fcγ receptor binding was abolished while FcRn and target binding were preserved, confirming successful Fc silencing without loss of functional integrity.; A scFv-Fc bispecific antibody targeting NKG2A and PD-1 was also transiently expressed. The molecule showed efficient heterodimerisation and simultaneous dual-target binding, with optimal assembly achieved through single-vector co-expression of both monomers.; These findings position Plant Molecular Farming as an integrated antibody engineering platform capable of producing functionally advanced, affordable immunotherapies and support its potential for decentralised biologics manufacturing and improved global access.

PF-05 Wednesday 13 May | 17:50-17:55

Optimizing the assembly and accumulation of SIgA in the endomembrane system of plants

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Topic: Plant-Made Biologics

The plant cellular machinery is well suited for the expression and assembly of SIgAs, but the overall yield and the proportion of fully assembled protein still vary. Expanding the physical dimensions of the ER generally supports a high protein production rate. This was confirmed in N. benthamiana by CRISPR/Cas-mediated enhancement of phosphatidylcholine biosynthesis, which resulted in a higher yield of fully assembled and functional SIgA in the mutant plant lines (Goritzer et al., 2025).; To find out whether and in which way the luminal volume and morphology of the ER influence interactions between antibody components and ER-resident host factors, we have started to establish MS-based identification of interacting cellular machinery, enabling a direct comparison between wild-type and mutant plants. This will help us identify host proteins as candidates for simultaneous overexpression alongside recombinant proteins, aiming to equip the enlarged ER for optimal function. ; In a complementary approach, we are investigating whether reprogramming the lytic vacuole can enhance overall SIgA yield. Although SIgA is primarily targeted for secretion, previous studies have indicated that a fraction is frequently mis-targeted to the lytic vacuole (LV), where it undergoes degradation. Since the extent of this mis-targeting is currently unknown, we are investigating whether reprogramming the LV can improve overall SIgA yield.

PF-06 Wednesday 13 May | 17:55-18:00

Multigenerational Expression and Purification of a Parathyroid Hormone-Fc Biologic in Transgenic Lettuce for Osteopenia Treatment

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Topic: Plant-Made Biologics

Osteopenia results from impaired bone remodeling that lowers bone density and increases fracture risk, particularly in aging populations and individuals exposed to microgravity. The biologically active region of the parathyroid hormone, PTH[1-34], is clinically approved for osteoporosis treatment. To extend its short circulating half-life, PTH was genetically fused to the fragment crystallizable (Fc) region of human IgG1 (PTH Fc).; Plant-based platforms offer scalable, economical biomanufacturing with reduced pathogen risk. Among these, Lactuca sativa (lettuce) is a promising host due to rapid growth, prior success producing therapeutics, and demonstrated spaceflight performance. This work aims to quantify PTH Fc accumulation in transgenic lettuce and evaluate a scalable Protein A-based downstream process.; Transgenic lines were generated by Agrobacterium-mediated transformation and screened across T1 - T3 generations by PCR, Western blot, and ELISA. Stable PTH-Fc transgene inheritance was confirmed through T3, with 100% of progeny in selected lines carrying the transgene. The lead line accumulated 0.3 - 1.2 mg PTH-Fc/kg fresh weight (0.004-0.018% TSP) in T2 plants. Crude plant extracts were concentrated and buffer-exchanged by tangential flow filtration, followed by Protein A affinity chromatography, yielding purified PTH-Fc at 0.85 mg/mL. These results demonstrate that transgenic lettuce supports scalable PTH-Fc production suitable for downstream protein characterization.

PF-07 Thursday 14 May | 18:20-18:25

More Than Carriers: Plant Viral Nanoparticles promote neuronal differentiation and growth guidance

M Ritter(1), Natalija Stojanovic (1), Simon Zschieschang (2), Johannes Grader (2), MHD Naeem Assasa (3), Eva Miriam Buhl (3), Andrea Coschiera (1), Stefan Schillberg (2, 4), Horst Fischer (1) and Juliane Schuphan (2)

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Topic: Viral Vectors & Nanoparticles

Despite advances in fibrous scaffolds for nerve repair, most platforms fail to integrate biochemical and physical cues essential for neuronal differentiation and neurite extension in vitro. Hey also lack precise control over growth factor presentation and spatial organization. Here, we introduce a multifunctional carrier based on potato virus X (PVX) nanoparticles that simultaneously delivers biochemical cues and directional guidance to neurons. Plant-produced, genetically engineered PVX is used to display high-density bioactive peptides derived from laminin and brain-derived neurotrophic factor, enabling targeted activation of neuronal membrane receptors. Neuronal differentiation and activation of associated signaling pathways are assessed using immunostaining, gene expression analysis, immunoprecipitation, and western blotting. Spatial guidance is achieved by aligning PVX nanoparticles using 3D-printed sacrificial materials to create an oriented microarchitecture, as confirmed by scanning electron microscopy. This integrated biochemical and structural strategy enables simultaneous pathway activation and neurite guidance within a single system. By combining biomimetic peptide display with plant-derived engineered PVX, this work establishes a versatile, scalable, and sustainable platform for precise control of neuronal differentiation and alignment, advancing biomimetic approaches in neural tissue engineering and regenerative medicine.

PF-08 Thursday 14 May | 18:25-18:30

Plant production of multivalent protein nanoparticle vaccines to prevent non-typhoidal Salmonella enterica colonization in poultry

CA Charron*(1)(2), S Shamriz (3), A Kaldis (1), CP Garnham (1)(4), MS Diarra (3), and R Menassa (1)(2)

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Topic: Viral Vectors & Nanoparticles

Non-typhoidal Salmonella enterica serovars (NTS) are major causes of gastrointestinal disease in humans, largely due to the consumption of contaminated poultry products. While vaccination is the most effective strategy to reduce NTS colonization in poultry, current vaccines have limitations regarding safety and efficacy, highlighting the need for improved alternatives. This project investigates plant-produced, self-assembling protein nanoparticles as a novel multivalent vaccine approach for NTS. Antigenic epitopes from three outer membrane proteins were genetically fused to the encapsulin monomer to achieve multivalency at three levels: multiple antigenic targets, multiple epitopes per antigen, and multiple copies of each epitope on the nanoparticle. The multivalent constructs were transiently expressed in Nicotiana benthamiana and successfully assembled into nanoparticles, despite lower yield than single-epitope versions. Further analysis revealed that epitope identity, rather than the number of modifications, primarily influenced protein stability and accumulation. A chicken immunization and NTS challenge study demonstrated production of NTS- and antigen-specific antibodies. Ongoing experiments will assess effects on NTS colonization, vaccine optimization, and strategies for oral delivery. Overall, these findings support the feasibility of multivalent, plant-produced encapsulin nanoparticles as promising next-generation NTS vaccines.

PF-09 Thursday 14 May | 18:30-18:35

A multilayer strategy for stable integration and inducible control of TMV-based biofactories

Marta Rodriguez-Rodriguez*, Marta Vazquez-Vilar and Diego Orzaez

Instituto de Biologia Molecular y Celular de Plantas, CSIC-UPV, Valencia, Spain

Topic: Viral Vectors & Nanoparticles

Viral vectors like the Tobacco Mosaic Virus (TMV) are efficient tools for recombinant protein production in plants, often yielding higher accumulation levels than standard constitutive expression systems. While current industrial practices rely on transient expression via agroinfiltration, scaling this process presents logistical and economic challenges. The genomic integration of viral replicons offers a potential solution; however, the autonomous replication of these vectors typically leads to significant cellular stress, hindering the regeneration of transgenic plants. To address these challenges, we developed a multilayered regulatory strategy to ensure tight control over integrated TMV replicons. First, we implemented an autonomous bioluminescence reporter to monitor viral kinetics with higher sensitivity. To keep a repressed state during plant development, we designed a circuit combining transcriptional regulation via copper-inducible promoters with a recombinase-mediated DNA inversion switch. In this configuration, the RNA-dependent RNA polymerase (RdRp) is maintained in an antisense orientation, preventing replication until the specific induction signal. This approach successfully mitigates viral leakiness in transient assays, being a promising approach to regenerate stable plants carrying the replicon. This system would offer a robust, scalable, and secure platform, paving the way for the next generation of industrial molecular farming.

PF-10 Thursday 14 May | 18:35-18:40

Scaled-Up Woody Plant Cell Suspension Cultures as Versatile Platforms for Extracellular Vesicles and Enzyme Production

B Casimiro (1), M Rito (1), S Caeiro (2), P Rosa (2), C Azevedo (2), P Verissimo (3), LF Rojas (4), JM Canhoto (1), S Correia*(1)

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Topic: Scaling Up Plant-Based Production

The increasing demand for sustainable, controllable, and scalable production of plant bioactives highlights the need for diverse in vitro metabolite platforms tailored to specific compound classes. Plant cell suspension cultures offer eco-friendly, reproducible systems that avoid whole-plant variability and allow controlled biosynthesis and recovery of valuable molecules. In this study, we established and optimised two complementary suspension-based platforms from woody species to produce different bioactives. Callus-derived cultures from Poplar (Populus tremula × P. alba) were developed for extracellular vesicle (EV) production, achieving isolation yields of 2.5×10^{1}■ vesicles per gram of biomass. RNA-loaded EVs successfully delivered cargo into Botrytis cinerea hyphae, demonstrating their potential as natural nanocarriers for plant protection and biotechnology. Simultaneously, tamarillo (Solanum betaceum) cell suspensions were optimised as enzyme-producing bioreactors through medium adjustments and elicitation. A 3% (w/v) sucrose medium enhanced growth and protein output, while casein hydrolysate, yeast extract, and chitosan selectively stimulated hydrolytic and glycosidase activities. Overall, these results show that tailored, scaled-up woody plant cell-suspension systems can act as versatile biofactories for vesicles, enzymes, and other high-value metabolites, thereby supporting biotechnological applications.

PF-11 Friday 15 May | 12:20-12:26

A genetic framework of genes involved in callus formation and organogenesis

Marcos Egea-Cortines

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Topic: Smart Chassis

Lavender hybrids such as Lavandula × intermedia ‘Budrovka' are increasingly explored as flexible upstream systems for plant molecular farming due to their well-characterized monoterpenoid and phenolic pathways and their reliable response to in vitro manipulation. Yet, long-term quantitative descriptions of callus biomass development remain limited. We monitored callus formation from root and stem explants in 3-week intervals across a 15-week cultivation period to generate a structured dataset suitable for predictive modelling. Five algorithms (Beta Regression, GAM, Random Forest, SVR, XGBoost) were trained on mean response data to estimate callus weight and induction efficiency. XGBoost achieved the highest accuracy for biomass prediction (R2 = 0.9373; RMSE 1.76 g), and feature-importance profiles consistently identified explant origin, culture duration and auxin-cytokinin regime as dominant contributors to callus performance. NSGA-II multi-objective optimization, guided by GAM and XGBoost outputs, indicated that 15-week root-derived calli grown under 0.5 mg/L 2,4-D and 2 mg/L BAP represent the highest-yielding configuration. The study provides a data-resolved view of long-term callus behavior in hybrid lavender and a modelling-based approach for rational tuning of biomass production in plant-based bioprocesses.; This work was supported by project FPPV-31-2026.

PF-12 Friday 15 May | 12:26-12:32

Calnexin and calreticulin ER lectins chaperones in N-glycosylation pathway of proteins: One step to lift the quality control bottleneck for production of glycoproteins

(Presenting author)Jules Delasalle(1,2), Narimane Mati-Baouche(1) *, Muriel Bardor(1)*, François Chaumont(2)* and Catherine Navarre(2)*

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Topic: Smart Chassis

Increasing demand for biologics, which are complex and glycosylated proteins, has driven the development of efficient sustainable bioproduction systems, like the microalgae Chlamydomonas reinhardtii and plant cell suspensions of Nicotiana tabacum BY-2, as alternatives to mammalian cell platforms. These systems offer: lower production costs, reduced contamination risks, and easy genome editing. N-glycosylation is well known to be a critical quality attribute affecting protein stability, activity and functions (Strasser et al., 2021). In the endoplasmic reticulum (ER), the lectin chaperones calreticulin (CRT) and calnexin (CNX) ensure proper folding of glycoproteins before Golgi transit (Aebi, 2013).; This study aims to: i) functionally characterize CRT and CNX in N. tabacum BY-2 and C. reinhardtii by generating KO cell lines. Phenotypic characterization: growth assays, RNA-seq, and glycan analyses are in progress. ii) improve recombinant glycoprotein production yield by overexpressing human, plant, and microalgal CRT/CNX variants together with a green-fluorescent erythropoietin fusion protein (EPO-Clover). Transient expression in Nicotiana benthamiana leaves revealed that co-expression of human CNX and CRT resulted in a 7-fold increase EPO-Clover accumulation. Stable transgenic N. tabacum BY-2 and C. reinhardtii lines were also generated, but differences in EPO-Clover fluorescence ratios were less pronounced under stable expression compared to transient systems.

PF-13 Friday 15 May | 12:32-12:38

Inhibition of transgene silencing in Nicotiana tabacum BY-2 cells via RDR genes inactivation and targeted demethylation

ROMANE VANDEVELDE (1), Laurent Bouhon (1), François Chaumont (1), Catherine Navarre (1)

(1) UCLouvain

Topic: Smart Chassis

Nicotiana tabacum Bright Yellow 2 (BY-2) cell suspensions are established as a platform for producing recombinant complex glycoproteins. However, this system typically yields limited amounts of protein, even when strong constitutive promoters are used. One limitation arises from RNA silencing processes, such as transcriptional gene silencing (TGS) and post-transcriptional gene silencing (PTGS), which prevent large accumulation of transgene RNA transcripts. To address this challenge, our research project aims at developing strategies to counteract the production of transgene dsRNAs involved in silencing as well as to relieve DNA methylation specifically established in the 35S promoter sequence. First, RNA-dependent RNA polymerase (RDR) genes involved in transgene silencing (RDR1, RDR2 and RDR6) were knocked-out in BY-2 cells using CRISPR/Cas9 technology. Next, BY-2 cell lines (WT, RDR2-KO, RDR1/6-KO and RDR1/2/6-KO backgrounds) expressing the dCas9-SunTag/TET1 system are currently being generated. This system recruits multiple copies of the catalytic domain of the human demethylase TET1 at a specific genome locus via a nuclease-deactivated Cas9 protein fused to repetitive GCN4 peptide epitopes, as previously reported. In our case, we will target demethylation of the 35S promoter controlling Gus reporter gene expression. We expect that this approach will significantly improve recombinant protein yields in transgenic BY-2 cells.

PF-14 Friday 15 May | 12:38-12:44

Biofortification of S. lycopersicum through the fungal bioluminescence pathway to enhance hispidin synthesis

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(1) Instituto de biologia molecular y celular de plantas (UPV-CSIC)

Topic: Metabolic Pathway Engineering

Plant biofortification is a cost-effective strategy for delivering health-promoting metabolites to populations with limited dietary diversity. While antioxidants like flavonoids are common targets, the potential of styrylpyrone-type compounds like hispidin--noted for antiviral, antidiabetic, and antioxidant properties--remains under-explored. In this study, we generated transgenic S. lycopersicum var. Micro-Tom lines constitutively expressing the Fungal Bioluminescence Pathway (FBP). This pathway synthesizes hispidin from caffeic acid via a polyketide synthase (HispS) and subsequently converts it into luciferin to produce autonomous light. We evaluated bioluminescence across growth stages and fruit ripening: immature green (IG), mature green (MG), and breaker plus seven (B+7). Our results show continuous, autonomous bioluminescence, notably concentrated in the fruit as a metabolic sink. LC-MS confirmed stable hispidin synthesis throughout all stages, correlating with strong HispS expression. This work demonstrates that natural caffeic acid levels in tomatoes are sufficient to sustain the FBP and accumulate hispidin. While bioluminescence serves as a reliable in vivo indicator of pathway activity, it does not correlate linearly with total hispidin content. These findings position hispidin as a promising target for metabolic engineering and highlight the FBP as a powerful tool for monitoring metabolite production in functional crops

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PF-15 Friday 15 May | 12:44-12:50

Characterization of Nicotiana-derived signal peptides and promoters for molecular farming

Javier Sanchez Vicente

(1), (4): Madeinplant SL. (2),(3): IBMCP (UPV-CSIC)

Topic: Molecular Farming Marketplace

Plant-based expression systems are widely used in molecular farming, but their efficiency strongly depends on the regulatory and targeting elements employed. This study aimed to characterize a panel of signal peptides and promoters from Nicotiana benthamiana and Nicotiana tabacum for recombinant protein production.; Several endogenous plant proteins that accumulated in the apoplast in response to agroinfiltration were identified, and their associated signal peptides and promoters were selected for evaluation. Their performance was tested in transient expression assays using different recombinant proteins and compared with commonly used signal peptides and promoters in plant biotechnology.; Our results suggest substantial variability in performance, particularly for signal peptides, which appears to be protein-dependent. One promoter showed especially promising results across constructs. These findings contribute to expanding the molecular toolbox available for plant molecular farming and support the use of endogenous Nicotiana regulatory elements as valuable alternatives for recombinant protein expression.

POSTER PRESENTATIONS

Poster sessions: Wednesday 13 May 20:00 & Thursday 14 May 18:40

P-01 Wed 13 & Thu 14 May

Floating Biofactories for Sustainable Oil Production

Amanda de Santana Lopes

Johannes Gutenberg-Universitat Mainz

Topic: Metabolic Pathway Engineering for High-Value Compounds

Duckweeds are floating aquatic plants recognized as the smallest flowering plants. They are particularly notable for their ease of cultivation and rapid growth. This latter trait makes them especially attractive for biotechnological applications, where fast biomass accumulation can be leveraged for the production of high value compounds. Recently, we published a detailed method for the reproducible genetic engineering of Spirodela polyrhiza (giant duckweed) to support duckweed research and expand its potential in biotechnology. One of our current projects focuses on engineering the triacylglycerol (TAG) metabolism of S. polyrhiza. Our long term goal is to develop a sustainable, land independent production system capable of achieving competitive oil yields. We first evaluated the heterologous expression of TAG-biosynthesis genes, including the transcription factors LEC2 and WRI1 and the enzymes DGAT1 and DGAT2. Among these, WRI1 and DGAT1 most effectively increased TAG accumulation in transformed S. polyrhiza. Then, to further enhance TAG levels, we combined WRI1, DGAT1, and the oil-body protein Oleosin in a single construct. Selection of transformed lines is ongoing, and regenerated plants will soon allow assessment of this strategy. Next, we aim to complement the heterologous expression strategy with CRISPR activation and dsRNA-mediated knockdown of endogenous targets to fine-tune TAG synthesis and maximize oil accumulation in giant duckweed.

P-02 Wed 13 & Thu 14 May

DISCOVERY OF OXYRESVERATROL BIOSYNTHESIS PATHWAY IN MULBERRY (Morus alba L.) USING CELLS UNDER MeJA ELICITATION: KEY ROLE OF C2'Hs ACTING UPSTREAM OF STILBENE SYNTHASES

Antonio Santiago Pajuelo

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Topic: Metabolic Pathway Engineering for High-Value Compounds

The medicinal properties of mulberry (Morus alba L.) are partly attributed to stilbenoids, including resveratrol (R) and oxyresveratrol (OxyR). Natural extraction from bark and roots is unsustainable, making biochemical characterization challenging. While R biosynthesis is established, the OxyR pathway remains unclear. To investigate this, we established mulberry cell suspensions where methyl jasmonate (MeJA) and methyl-beta-cyclodextrins (MBCD) elicited high intra- and extracellular R and OxyR production. We conducted time-series elicitation integrating transcriptomics and proteomics, supported by a re-annotated long-read genome (Mal_HE_154X_1). Multi-omics analysis identified 768 differentially expressed genes and 712 differentially abundant proteins. These included 22 stilbene synthase (STS) genes and six p-coumaroyl-CoA 2'-hydroxylases (C2'Hs) co-expressed with R and OxyR accumulation. We hypothesized that the C2'H product, 2,4-cinnamoyl-CoA, rather than R, serves as the OxyR precursor in an STS-catalyzed reaction. Characterization of a recombinant C2'H isoform confirmed conversion of p-coumaroyl-CoA into 2,4-cinnamoyl-CoA. Agroinfiltration in N. benthamiana and grapevine cells expressing MaC2'H and STS led to OxyR accumulation. These results conclude that OxyR biosynthesis in mulberry is mediated by C2'Hs acting upstream of STS.

P-03 Wed 13 & Thu 14 May

CRISPR Editing of a Light-Responsive Pathway Unlocks Polyphenol Biosynthesis in Annurca Apple

Carmen Laezza

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Understanding how environmental signals modulate secondary metabolism remains a key challenge in plant biology. In this study, we combined CRISPR-Cas9 gene editing with callus culture systems to investigate the genetic and environmental regulation of the dihydrochalcone and flavonoid pathways in Malus domestica cv. Annurca. This traditional apple cultivar from Southern Italy is renowned for its rich polyphenolic profile, making it an ideal model to study light-dependent metabolic regulation. Peel-derived callus cultures provided a rapid and controllable platform for targeted gene manipulation and metabolic analysis. Following MdCHI gene knockout, both control and edited lines were grown under light and dark conditions to examine how light influences pathway activation. Expression profiling of key biosynthetic genes (Md4CL, MdCHS, MdCHI, MdUGT, and MdFLS) showed that light strongly induces their expression, causing pronounced differences in polyphenol and flavonoid accumulation, particularly in edited lines. These results demonstrate that combining gene editing and in vitro culture systems enables functional dissection of environmentally responsive biosynthetic networks and enhances our understanding of light-regulated secondary metabolism. Future RNA-seq analyses will identify transcription factors regulated by light and involved in activating polyphenol pathway genes in Annurca apple, providing new insights for targeted manipulation of secondary metabolism in fruit crops.

P-04 Wed 13 & Thu 14 May

Uncovering the Metabolic Fate of Vanillin in Plants for Vanilla Aroma production

Giulia Marchisella

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Vanillin is the predominant odour component of natural vanilla, a complex aromatic bouquet comprising more than 200 compounds. Natural vanilla is obtained from Vanilla planifolia pods through a time-consuming process, whereas commercially available vanilla largely consists of pure vanillin produced by chemical synthesis or microbial fermentation. However, vanillin alone does not reproduce the full complexity of the natural bouquet.; In plants, the enzymatic route leading to vanillin formation and its metabolic fate remain unclear. Elucidating vanillin metabolism in plant cells is essential to identify upstream and downstream constraints affecting its accumulation. This project aims to investigate plant metabolic responses associated with vanillin as a foundational step toward the long-term reconstruction of the natural vanilla bouquet through plant-based metabolic engineering.; Accordingly, callus cultures from different plant species were fed with vanillin and analysed using a metabolomic approach. UPLC-MS profiling revealed extensive and species-specific conversion of vanillin into secondary metabolites, indicating the presence of endogenous transformation pathways. These results provide insights relevant to host selection and pathway engineering, supporting plants as promising platforms for vanilla aroma biosynthesis.

P-05 Wed 13 & Thu 14 May

INTERPLAY BETWEEN LIGHT SIGNALING AND PHYTOHORMONE ELICITATION IN THE REGULATION OF ANTHOCYANIN, STILBENOID, AND TERPENE BIOSYNTHESIS IN GRAPEVINE CELL CULTURES

Jone Echeverria Alberdi

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Plants produce specialized metabolites of high industrial value, yet the signalling cues regulating their biosynthesis remain complex. This study uses Vitis vinifera cell cultures to investigate the regulatory networks driving different specialized metabolic pathways. We evaluated the impact of jasmonic acid (JA, MeJA), cyclodextrins (CD) and abscisic acid (ABA) treatments under light and dark conditions. Anthocyanin accumulation was strongly light-dependent. While JA+CD elicitation significantly enhanced levels in the dark, light exposure potentially masked this effect. Conversely, ABA promoted higher anthocyanin levels regardless of light, indicating distinct regulatory mechanisms. Resveratrol, undetected in controls, was induced by JA+CD in both regimes, with higher levels in light-grown cultures, suggesting interaction between light signalling and stilbenoid regulation. Terpene accumulation was also responsive to JA-based elicitation, but not to ABA. To identify transcriptional drivers of these responses, transcriptomic data from 24 h and 4 day elicited cells was analyzed. This identified known and uncharacterized factors, including MYBA, MYC2, MYB24, MYB14 and WRKY03. DAP-seq and dual-luciferase analyses corroborated their regulatory roles. Finally, a 24-hour elicitation time-course was conducted to help resolve the temporal hierarchy among these TFs. Together, these findings provide insight into plant specialized metabolism in grape cells.

P-06 Wed 13 & Thu 14 May

Valproic acid as an epigenetic modifier to improve the production of (non)protein valuable metabolites: early-stage results

Kateryna Lystvan

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Influencing the epigenome is one of the ways to enhance the production of valuable substances through modifications that alter chromatin structure and promoter accessibility, thereby modulating gene transcriptional activity. Altering the activity or the heterologous expression of epigenome-modifying enzymes can activate hitherto inactive genes or enhance the expression of low-activity ones, leading to the synthesis of new metabolites or the hyperaccumulation of target (non)recombinant products. There are successful examples of using such an approach in microorganisms. However, it remains poorly studied in plant biotechnology, despite some reports on the beneficial effect of histone deacetylase (HDAC) inhibitors on recombinant protein production. We aimed to fill this gap (at least partially) by initiating a comprehensive analysis of the effects of both epigenome-modifying small molecules and heterologously expressed genes to improve the production of (non)protein metabolites. Our early-stage results confirm this potential. While we previously demonstrated the significant impact of valproic acid (VA), a potent HDAC inhibitor, on the regeneration processes of some plant species, we have now also observed its beneficial effects on protein accumulation and activity. Specifically, an 10-15% increase in total protein content, alongside a robust (up to 2.5-fold) enhancement of antibacterial activity, has been observed in VA-treated recombinant colicin-producing N. tabacum plants.

P-07 Wed 13 & Thu 14 May

VmMYBA1 transcription factor is a strong inducer of anthocyanin biosynthesis in Nicotiana cells

Katja Karppinen

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Bilberry (Vaccinium myrtillus L.) fruits accumulate high concentration of anthocyanins, especially delphinidins, which are associated with health-benefits. Anthocyanins are produced via the flavonoid biosynthetic pathway, which is regulated by the MBW regulatory complex, including R2R3 MYB transcription factor (TF) that is considered as the main regulator in the complex. Among 18 flavonoid biosynthesis-regulating R2R3 MYB genes identified in bilberry genome, VmMYBA1 was associated with anthocyanin biosynthesis in fruit. Transient overexpression of VmMYBA1 in Nicotiana benthamiana leaves by Agrobacterium infiltration led to the induction of expression of anthocyanin biosynthesis genes and accumulation of delphinidin 3-rutinoside. VmMYBA1 was verified to activate the promoters of F3'5'H, DFR, ANS and UFGT in the flavonoid pathway. When VmMYBA1 was cloned into a pEAQ-HT-DEST3 expression vector and transiently overexpressed in N. tabacum, a high accumulation of anthocyanins was observed. Our results indicate that VmMYBA1 can produce high anthocyanin concentration in Nicotiana cells without the need for co-infiltration of other TFs of the MBW complex or structural genes, such as F3'5'H to derive delphinidin production. VmMYBA1 is an excellent candidate for metabolic engineering towards high anthocyanin/delphinidin production in tobacco cells, which represents a sustainable alternative to conventional anthocyanin production platforms for use in industrial and medical applications.

P-08 Wed 13 & Thu 14 May

Cultured Plant Cells as Versatile Biofactories for Carotenoids and Functional Proteins

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Plant cell culture systems offer a sustainable and controllable platform for producing high-value biomolecules. Here we provide an overview of our work on the use of cultured plant cells as versatile biofactories, including metabolic engineering and the production of recombinant functional proteins. We established a plant-based platform using metabolically engineered tobacco BY-2 and Medicago truncatula A17 cell suspension cultures to produce ketocarotenoids, astaxanthin and canthaxanthin, by heterologous expression of key enzymes of the biosynthetic pathway. The intracellular localization of the accumulated carotenoids was investigated, and preliminary scale-up approaches were also explored. In parallel, plant cell systems were developed for the production of functional recombinant proteins, including health-related proteins such as the large complex Spike viral glycoprotein, as well as food-relevant proteins such as sweet proteins like thaumatin. Overall, these studies demonstrate the flexibility of plant cell cultures in supporting distinct biosynthetic objectives within a unified technological framework. Our results highlight the potential of cultured plant cells in health and food biotechnology, including emerging cellular agriculture strategies, reinforcing their role as adaptable, safe production platforms for next-generation bio-based products.

P-09 Wed 13 & Thu 14 May

Biomimicking mammalian ER in plants

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

The endoplasmic reticulum (ER) is central to protein biosynthesis and folding, yet plant ER architecture is predominantly tubular, unlike the sheet-rich ER of mammalian secretory cells, which is optimized for high biosynthetic output. This project aims to reprogram plant ER morphology using synthetic and conserved ER-shaping proteins to induce mammalian-like ER sheets and to explore the effects on biosynthetic capacity. Arabidopsis thaliana ER-shaping proteins LNP1 and the reticulons RTNLB10 and RTNLB13 have been selected for transient overexpression in Nicotiana benthamiana, together with selected Nicotiana homologs, to assess their functional conservation in ER sheet stabilization, curvature regulation, and network organization. In addition, the mammalian ER scaffold protein CLIMP-63 will be heterologously expressed in plant cells to assess its ability to stabilize sheet-like ER domains in plant cells. We are currently assessing all constructs via transient agroinfiltration, allowing for rapid functional screening. Based on these outcomes, CRISPR/Cas9-based genome editing will be considered for precise and combinatorial ER engineering in N. benthamiana. ER remodeling and spatial organization will be analyzed using advanced microscopy. Potential effects on protein folding and post-translational processing will be explored to link ER morphology with biosynthetic performance and its relevance for plant-based molecular farming.

P-10 Wed 13 & Thu 14 May

Expression of Rat and Rabbit Hepatitis E Virus Capsid Proteins in Nicotiana benthamiana

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Rat hepatitis E virus (rat HEV) and rabbit hepatitis E virus (raHEV) have been isolated from their respective animal reservoirs, rats and rabbits, and accumulating evidence indicates their potential for zoonotic transmission to humans. Nicotiana benthamiana has emerged as an efficient and scalable plant-based expression system for the production of viral proteins and virus-like particles (VLPs). However, the expression of rat HEV and rabbit HEV capsid (ORF2) proteins capable of assembling into VLPs in N. benthamiana has not yet been reported. In this study, truncated ORF2 capsid proteins derived from rat and rabbit HEV were evaluated for expression in N. benthamiana. Two rat HEV ORF2 constructs and one rabbit HEV ORF2 construct, containing N- and C-terminal deletions (corresponding to amino acid positions rat HEV 357-614 and 102-614, and rabbit HEV 112-507), were cloned into the pEAQ-HT expression vector and transiently expressed in N. benthamiana leaves. The successful expression of these truncated capsid proteins provides a basis for assessing their capacity to assemble into VLPs and supports the development of plant-based platforms for the production of HEV structural proteins. This work establishes a foundation for further characterization of rat and rabbit HEV VLPs formation in plants and highlights the potential of N. benthamiana as a versatile system for studying zoonotic HEV and for developing diagnostic or vaccine-related reagents.;

P-11 Wed 13 & Thu 14 May

Using CRISPR/ Cas9-mediated gene editing strategy to produce of the oral shrimp recombinant vaccine in green microalgae

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Microalgae have high growth rate, ease of cultivation, low costs and metabolic pathways that are similar to those of higher plants, leading to the same post-transcriptional and post-translational modifications. since micro-algae are edible and contain nutritionally valuable substances such as vitamins, proteins, and fatty acids, they can be used directly without isolation and purifying recombinant proteins to consume animals. Among various anti-WSSV strategies, envelope proteins are preliminary target for the development of vaccines. Because, envelope structural proteins " like VP28" play a crucial role in viral infections, and are often involved with viral entry, assembly and budding processes. In this study, the coding sequence of VP28 was transferred to the pChlamy4 vector; then pChlamy4-VP28 electroporated to C. reinhardtii.(UVM4 strain). The expression of vp28 protein was confirmed using SDS-PAGE and WB. Overall, the expression level of vp28 protein in the present study was low and had limited commercial and practical aspects. Therefore, in the continuation of the current research, an attempt is made to first Selection of the desired genomic locations for insertion of target Gene; in the genome of the microalgae including repetitive sequences and other suitable gene insertion sites, and then, using modern molecular techniques including CRISPR/CAS system, specifically insert the target gene into multiple sites with multiple repetitions to improve the expression level.

P-12 Wed 13 & Thu 14 May

Development of an efficient purification process for monoclonal antibodies transiently expressed in Plant Cell Packs

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Topic: Plant-Made Biologics – From Antibodies to Advanced Therapeutics

The Plant Cell Pack (PCP) technology as transient expression system for plant cells has recently been scaled for the production of first-wave therapeutics such as needed in case of emerging diseases or personalized medicine, in which monoclonal antibodies (mAbs) as vaccine are of particular interest. Using an anti-malaria and an anti-cancer mAb, we therefore established a high-yield mAb purification protocol for the PCPs. Comparing the default combination of mechanical blending and high-pressure homogenization with mechanical blending alone revealed that homogenization increased the target protein yield by less than 10%, while skipping high-pressure homogenization step allowed the biomass-to-buffer ratio to be reduced from 5:1 to 2:1, improving process efficiency substantially. We further identified three depth-filter setups that provided recoveries in the range of 80% but achieved a strong reduction in turbidity <20 NTU, thereby increasing the capacity of the sterile filter. Finally, by optimizing the elution conditions during Protein A chromatography, we obtained almost completely monomeric mAbs with <3% aggregates and increased the recovery from 50% to over 90%. In parallel, we established quality control analytics including screening of mAb aggregation by dynamic light scattering (DLS) and mAb binding kinetics by surface plasmon resonance (SPR) as prerequisite for a Good Manufacturing Compliant (GMP) process which is currently discussed with the regulatory authorities.

P-13 Wed 13 & Thu 14 May

ENGINEERING SECRETORY ANTIBODIES FOR MUCUSAL IMMUNITY: ADVANCES, CHALLENGES AND EMERGING APPLICATIONS IN NON-TRADITIONAL EXPRESSION PLATFORMS

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

The pharmaceutical industry is a fast-growing industry, and as a result, there is an increased demand for alternative production systems. Sectors such as antibody-based biopharmaceuticals are among the fastest growing sectors in the pharmaceutical industry. Plants such as Nicotiana benthamiana can produce enormous quantities of different recombinant proteins. Secretory antibodies (SIgA) are found in mucosal areas such as the gastrointestinal and respiratory tract and are essential in regulating and protecting intestinal, respiratory, and urogenital mucosal epithelia by limiting microorganism invasion/access. Given that N. gonorrhoeae primarily invades the mucosal areas of the urogenital tract, SIgA-based therapeutics could offer a targeted approach towards preventing bacterial adherence and infection. The scientific community's knowledge on plant-made N. gonorrhoeae-targeting secretory antibody is limited. In this review, our basic understanding of engineering secretory antibodies for mucosal immunity is described along with advances, challenges and emerging applications in non-traditional expression platforms. This study will provide novel insights into its feasibility as an alternative therapeutic strategy due to the limited research on plant-produced SIgA's against N. gonorrhoeae.

P-14 Wed 13 & Thu 14 May

Differential Host Responses to Full-Length and Mature Serine Proteases during Transient Expression in Nicotiana benthamiana

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Plant molecular farming enables rapid and scalable production of recombinant biopharmaceuticals; however, expression of proteolytic enzymes remains challenging due to host toxicity and tissue necrosis. In this study, we investigated the impact of protease maturation state on plant host responses using transient expression in Nicotiana benthamiana. Building on our published findings with nattokinase, we compared the expression of full-length zymogen forms and corresponding mature serine proteases lacking pro-domains.; Agroinfiltration of full-length constructs consistently triggered rapid leaf necrosis within 3-5 days post-infiltration, accompanied by reduced protein accumulation and compromised tissue integrity. In contrast, expression of mature protease forms resulted in delayed or minimal necrosis, higher soluble protein yield, and preserved leaf morphology. Functional assays confirmed that mature enzymes retained robust proteolytic activity, whereas full-length forms exhibited limited recoverable activity due to early tissue collapse.; These observations suggest that pro-domains in plant-expressed serine proteases may interfere with host cellular processes or activate plant immune responses, leading to hypersensitive-like cell death. Our results indicate that strategic removal of pro-peptides can mitigate host toxicity while preserving enzymatic function, offering a generalizable design principle for expressing bioactive proteases in plants.

P-15 Wed 13 & Thu 14 May

Expression of a recombinant human Granzyme B (GrB) in Nicotiana benthamiana plants for tumour-targeting applications

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

This work aims to produce in plants an antibody-drug conjugate to specifically target glioblastoma tumors, inducing cell death. To this aim the monoclonal antibody H10 specifically recognizing the C domain of Tenascin-C, a tumor marker of the extracellular matrix was fused to granzyme B (GrB), a serine protease normally found in the granules of natural killer cells and cytotoxic T cells which mediates apoptosis in target cells. In this work, we have verified the possibility of producing functionally active GrB in plants. To this aim, the human GrB fused to an antibody Fc portion (GrB-Fc) in order to stabilize the protein and to facilitate purification. The recombinant enzyme was produced in plants by Agroinfiltration in its inactive form with an N-terminal DDDDK peptide and purified by protein A affinity chromatography at levels of 4mg/Kg FW. The enzyme could be easily activated by enterokinase treatment and in vitro experiments, using U87 glioblastoma cells, revealed its biological activity. Furthermore, the GrB was directly fused to the recombinant scFvH10 antibody. This antibody-drug conjugate (GrB-scFvH10) was expressed and successfully purified from plant agroinfiltrated leaves at levels of 30 mg/Kg. Preliminary in vitro tests show its functional activity on glioblastoma cells. Future experiments will explore the possibility to use GrB-scFvH10 in antibody-mediated drug delivery applications in mouse models against solid tumors expressing the C-Domain of Tenascin C.

P-16 Wed 13 & Thu 14 May

Plant production of BP26 and SP39 Brucella suis antigens fused to swine Fc for potential vaccinal applications

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Brucella suis is the causative agent of porcine brucellosis, a globally important zoonosis with a significant impact on animal health and food safety. There is no vaccine available highlighting the need to explore novel antigenic products for the development of new vaccines and diagnostic tests. In the present work, two immunogenic B. suis periplasmic proteins (BP26 and SP39), were selected to be produced in N. benthamiana plants. Both proteins were transiently expressed in plants as fusion with the Fc portion of a porcine IgG1 (BP26-Fcp and SP39-Fcp) with the aim of potentiating their immunogenic properties, increasing expression stability and facilitating purification. About 50 mg/kg of leaf fresh weight of each protein were recovered and biochemical characterization showed that SP39-Fcp forms dimers while BP26-Fcp assembles into multimers. Transmission electron microscopy and DLS analysis revealed the ability of BP26-Fcp to form nanobarrel-like structures, as bacterial BP26, with a hydrodynamic diameter of about 15 nm. Results also showed that the Fc moiety maintains the ability to specifically interact with porcine FcγRIIIa, indicating that this type of antigen design may result in the efficient targeted delivery to immune cells expressing this receptor type. We demonstrated that plants are an efficient system to produce B. suis antigens fused to a porcine Fc for possible vaccinal applications and that BP26-Fcp nanobarrels may have a potential use in nanotechnology.

P-17 Wed 13 & Thu 14 May

Targeted delivery of a plant-derived monoclonal antibody using pulsed electric fields in a 3D glioblastoma model.

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

The objective of this project is the targeted treatment of solid tumors using electroporation. The strategy involves the plant-production of the monoclonal antibody H10 (mAb H10) specifically recognizing the C domain of Tenascin-C (C-TNC), a tumor marker present in the extracellular matrix of solid tumors such as glioblastoma. This aggressive brain tumor is characterized by the overexpression of C-TNC which takes part in proliferation and pro-angiogenic effects within the tumor microenvironment. The plant-purified antibody was conjugated with a fluorescent molecule to conduct in vitro studies. The targeting efficacy of mAb H10 was tested on a glioblastoma spheroid model (able to mimick the tumor architecture) and different electroporation protocols were assayed to enhance the antibody's ability to penetrate the neurospheres. Electroporation was shown to facilitate the entry of mAb H10 into the spheroid matrix, resulting in a more homogeneous fluorescence within the treated neurospheres compared to non-electroporated controls. Moreover, confocal microscopy analysis evidenced the accumulation of the antibody in the extracellular matrix within the spheroids. These results demonstrate that electroporation-assisted delivery of mAb H10 into glioblastoma spheroids is a promising method for enhancing targeted therapy. Future research should focus on developing non-invasive electroporation devices and evaluating the therapeutic potential of this molecule.

P-18 Wed 13 & Thu 14 May

Artemisia at flowering: optimizing in planta accumulation and in vitro evaluation of secondary metabolites with potential anticancer activity

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Aromatic plants often synthesize specialized metabolites at very low levels within specific structures or at precise developmental stages, hindering their large-scale exploitation. This is the case of Artemisia annua, used in Traditional Chinese Medicine and recommended by WHO for malaria treatment. Its active compound Artemisinin (AN) accumulates only in glandular trichomes and mainly at flowering, making its production difficult in the quantities needed for therapeutic results. Several approaches have been explored to enhance AN production, including the use of bio-stimulants that accelerate the vegetative to reproductive phase change and double trichome density. Analyses of plants treated with specific Plant Growth Regulators led to the identification of candidate genes likely involved in the control of the floral transition. Our research focuses on the characterization of these regulatory genes to better understand how photoperiodic flowering is linked to AN biosynthesis. Beyond AN, A. annua synthesizes additional metabolites relevant to human health. We first optimized “quick & cheap” methods to extract terpenoids and flavonoids from dried leaves and floral buds. We then tested the anticarcinogenic capacity of these extracts using 3D models - recognized as standard for the tumorigenic potential of malignant cells. Some phytocomplexes markedly reduce spheroid formation in vitro, suggesting a great potential in reducing tumor cell reawakening.;

P-19 Wed 13 & Thu 14 May

Production of a virus like particle for the vaccination of East Coast Fever using antigen p67c: recombinant protein expression Agrobacterium tumefaciens, and production in Nicotiana benthamiana and Chlamydomonas

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

The tick borne Theileria parasite poses a great risk to cattle within sub-Saharan Africa, as it causes East Coast Fever, which can lead toillness and death in cattle. Global warming is causing extended habitat range and prolonged active-periods of Theileria. The pressing issue currently is the lack of a viable, accessible and affordable vaccine. We propose to express a virus-like particle (VLP) of the hepatitis B core antigen (HBcAg) which displays the Theileria surface antigen p67c. A recombinant VLP was designed for transient expression in Nicotiana benthamiana, and additional production in the alga Chlamydomonas. The p67c antigen alone was also expressed in plants. Expression, purification, and characterisation (structural and immunogenic) of the vaccine candidates will be presented.

P-20 Wed 13 & Thu 14 May

High-yield, plant-based production of an antimicrobial peptide with potent activity in a mouse model

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Plants offer a promising chassis for the large-scale, cost-effective production of diverse therapeutics, including antimicrobial peptides (AMPs). However, key advances will reduce production costs, including simplifying the downstream processing and purification steps. Here, using Nicotiana benthamiana plants, we present an improved modular design that enables AMPs to be secreted via the endomembrane system and sequestered in an extracellular compartment, the apoplast. Additionally, we translationally fused an AMP to a mutated small ubiquitin-like modifier sequence, thereby enhancing peptide yield and solubilizing the peptide with minimal aggregation and reduced occurrence of necrotic lesions in the plant. This strategy resulted in substantial peptide accumulation, reaching around 2.9 mg AMP per 20 g fresh weight of leaf tissue. Furthermore, the purified AMP demonstrated low collateral toxicity in primary human skin cells, killed pathogenic bacteria by permeabilizing the membrane, and exhibited anti- infective efficacy in a preclinical mouse (Mus musculus) model system, reducing bacterial loads by up to three orders of magnitude. A base-case techno-economic analysis demonstrated the economic advantages and scalability of our plant-based platform. We envision that our work can establish plants as efficient bioreactors for producing preclinical-grade AMPs at a commercial scale, with the potential for clinical applications.

P-21 Wed 13 & Thu 14 May

Augmenting the Plant Cell Pack (PCP) transient expression technology to a fully automated screening platform for early-stage biopharmaceutical product development

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Biopharmaceuticals, particularly monoclonal antibodies (mAbs), are the fastest-growing class of therapeutics for immune, infectious, and oncologic diseases inadequately addressed by small-molecule drugs. Their development increasingly relies on artificial intelligence (AI)-supported protein engineering, thereby driving the demand for high-throughput screening (HTS) platforms to identify optimal target variants and production conditions. In this context, plants provide an efficient screening platform as the transient transformation enables rapid protein expression without the need for stable cell line generation, and for screening purposes the transient transformation was recently transferred to the microtiter plate (MTP) format via the Plant Cell Pack (PCP) technology. However, despite the automation of PCP transformation itself, the throughput of the PCP technology is currently limited by manual cloning of expression vectors upstream and protein purification downstream. Using a robotic station, we developed a modular cloning workflow for vector assembly and transformation into Agrobacterium tumefaciens, and further established a purification workflow using magnetic beads. By integrating both workflows into the PCP platform, we created a fully automated, end-to-end system that streamlines and accelerates early-stage biopharmaceutical product development and enables the screening up to 384 different protein variants per day.

P-22 Wed 13 & Thu 14 May

Plant-based production of human IL-22-Fc with mammalian type glycans

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Recombinant human interleukin-22 (IL-22) is a promising therapeutic candidate due to its central role in immune regulation, tissue repair, and host defense against bacterial pathogens. IL-22 contains multiple N-glycosylation sites, and the attached N-glycans critically influence its pharmacokinetics and bioactivity, as specific glycan structures affect both protein half-life and binding to the IL-22 receptor, a prerequisite for IL-22-mediated signaling. Accordingly, precise monitoring and modulation of glycosylation are essential for optimizing the stability and functional performance of recombinant IL-22 variants. A CHO cell-derived recombinant IL-22-Fc fusion protein dimer has demonstrated an extended half-life compared with native recombinant IL-22 and has shown favorable safety and tolerability profiles in healthy subjects. However, IL-22-Fc is a heavily glycosylated protein and exhibits substantial glycan heterogeneity when produced in CHO cells, which may compromise product consistency, efficacy, and manufacturability. To address these limitations, we transiently expressed recombinant human IL-22-Fc via agroinfiltration in leaves of glycoengineered Nicotiana benthamiana. Mass spectrometry analysis of affinity-purified IL-22-Fc revealed efficient N-glycosylation site occupancy and the formation of mammalian-type N-glycans, highlighting the potential of the glycoengineered N. benthamiana-based expression platform for the controlled production of IL-22-based biologics.

P-23 Wed 13 & Thu 14 May

From Academic Spin off to GMP Manufacturing: The Evolution of Diamante Societa Benfit srl

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Translating academic research into clinically relevant therapies remains a major challenge in biomedical innovation. Virus nanoparticles (VNPs) represent a promising platform for targeted immunomodulatory therapeutics due to their structural versatility and scalable production. The combination of VNP with immunodominant peptide offers a novel strategy for antigen specific immunotherapies aimed at restoring immunological tolerance in autoimmune diseases. This poster describes the evolution of Diamante SB Srl, an innovative SME founded as a spin-off of the University of Verona, and its progression toward the industrial-scale development of a virus-like nanoparticle (VNP)-based, plant-produced therapeutic candidate for rheumatoid arthritis, as well as its approach to Phase I clinical trials. Diamante's platform is based on VNPs derived from Tomato Bushy Stunt Virus (TBSV), genetically modified to display the immunodominant peptide LIP1 associated with rheumatoid arthritis. Preclinical studies demonstrated the ability of the VNP based platform to induce antigen specific immunological tolerance. The transition to a new independent headquarters enabled the establishment of GMP compliant laboratories and workflows. Current activities focus on process optimization, quality control development, and regulatory alignment in preparation for GMP certification, supporting scalable production and clinical translation.

P-24 Wed 13 & Thu 14 May

Development of a molecular farming platform for high-level transient production of PCV2d capsid in Nicotiana benthamiana

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Porcine circovirus type 2 (PCV2), the prevalent PCV2d genotype, remains a major challenge for swine health due to antigenic divergence from current vaccines. This study aimed to establish a plant-based molecular farming platform for the production and preliminary evaluation of the PCV2d capsid (Cap) protein in Nicotiana benthamiana. A codon-optimized PCV2d cap gene with N-terminal truncation and subcellular targeting elements was transiently expressed in 4-5-week-old N. benthamiana plants via Agrobacterium-mediated infiltration. Leaves harvested at 3, 5, and 7 days post-infiltration were analyzed by SDS-PAGE and western blotting. Cap protein accumulated mainly as a ~27 kDa monomer, accompanied by higher-molecular-weight species (~48-63 kDa), with reproducibly highest accumulation at 7 days. Recombinant Cap protein was enriched by affinity-based purification and showed self-assembly behavior under controlled buffer conditions. Transmission electron microscopy confirmed spherical particles of approximately 17-22 nm, consistent with PCV2 virus-like particle (VLP) morphology. In a preliminary small-animal immunization study, plant-produced Cap protein induced detectable Cap-specific humoral immune responses without observable adverse effects. These results demonstrate that N. benthamiana supports efficient production of PCV2d Cap protein and its assembly into immunogenic VLPs, highlighting the potential of plant-based molecular farming for veterinary vaccine development.

P-25 Wed 13 & Thu 14 May

Evaluation of the transient expression of human activin receptor 2B Fc fusion protein in Nicotiana benthamiana

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Plants offer a versatile platform for bioproduct manufacturing, including Space, where they can provide both food and medicine. The potential therapeutic, activin receptor 2b Fc fusion protein, (ACVR2B-Fc) shows promise in counteracting muscle atrophy, reducing it in mice under microgravity and in genetic disease models. Here I present successful ACVR2B-Fc expression in tobacco, confirming viability of an alternative production system and setting up for a future transgenic platform in the space relevant plant, Lemna japonica. Using Agrobacterium tumefaciens-mediated infiltration of N. benthamiana, I evaluated four ACVR2B-Fc sequences, comparing two Fc sequences and two glycosylation patterns and ER retention. The recombinant ACVR2B-Fc protein was screened through a combination of bicinchoninic acid (BCA), SDS/PAGE, western blotting, and enzyme linked immunosorbent assay (ELISA) to evaluate expression levels. The Fc fusion of the protein also serves for use in purification and allows for streamlined protein evaluation. Transient expression was achieved for the two constructs that were retained within the ER, and expression of these two constructs was evaluated from 1 to 4 days post infiltration. Plants showed peak expression around day 3. Early analyses show ACVR2B-Fc as approximately 0.01-0.02%TSP(2-4mg/kg FW). This study establishes that ACVR2B-Fc can be expressed in plants and paves the way for future optimization in tobacco as well as transgenic expression in L. japonica.

P-26 Wed 13 & Thu 14 May

Targeting breast and ovarian cancer using a plant-produced scFv-anti-HER2 antibody

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Plant-based expression systems have emerged as attractive platforms to produce recombinant proteins and to allow protein engineering strategies. Here, we engineered and expressed a scFv anti-HER2 in Nicotiana benthamiana, incorporating the 2S2 signal peptide to promote efficient secretion into the apoplast. Agroinfiltration was performed with GV3101 Agrobacterium harboring the pK7WG2D_scFv-anti-HER2 plasmid. Apoplast redirection by the 2S2 signal peptide was confirmed by SDS-PAGE and Western blot in apoplastic fluid, even at low agroinfiltration densities. Positive samples were purified using His-tag affinity chromatography, diluted (200 to 0.001 µg/mL) and evaluated in HER2-positive breast (SK BR-3) and ovarian (SKOV-3) cancer cell models, using Trastuzumab as a reference therapeutic antibody. In the MTT assay, Trastuzumab reduced cell viability by less than 20%, keeping it above 80%, whereas plant-produced scFv-anti-HER2 decreased viability to about 50% at higher concentrations for both cell lines. Despite lacking Fc region, the scFv retained significant biological activity, exhibiting measurable cytotoxic effects with an IC■■ of approximately 37.36 µg/mL in SK BR-3 cells and 100 µg/mL in SKOV-3 cells. These findings demonstrate the effect of biologically active scFv-anti-HER2 fragments produced in plant-based platforms and highlight their potential as cost-effective and scalable alternatives for the development of next-generation targeted cancer therapeutics.

P-27 Wed 13 & Thu 14 May

The bacterial laccase CueO-MacHis can be produced at high yields in plant chloroplasts for large-scale industrial application

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Topic: Scaling Up Plant-Based Production

Laccases are versatile oxidoreductase enzymes present in bacteria, fungi, and plants, with numerous industrial applications, including the degradation of pharmaceutical micropollutants in wastewater. Among these, bacterial laccases are particularly attractive due to their high catalytic activity and remarkable stability across a wide range of temperatures and pH values. One such enzyme is the Escherichia coli copper efflux oxidase (CueO) fused to the adhesion promoting peptide Maqaque Histatin (MacHis) as immobilized enzymes proved to be most effective for waste water treatment. However, their practical application has been limited by low accumulation levels of only 25-50 mg/L when recombinantly expressed in E. coli or Pichia pastoris.; In this study, we demonstrate that the engineered CueO-MacHis can be produced at substantially higher levels of 250-320 mg/kg biomass in plant chloroplasts. Furthermore, we established a simplified and cost-effective purification strategy that leverages the enzyme's temperature and pH stability combined with either anion exchange chromatography (AEX) or tangential flow filtration (TFF).; This approach paves the way for the large-scale production of CueO-MacHis as a technical enzyme for industrial applications.

P-28 Wed 13 & Thu 14 May

Optimization of Long-Term Callus Biomass Production in Hybrid Lavender Using Machine Learning

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Topic: Scaling Up Plant-Based Production

Lavender hybrids such as Lavandula × intermedia ‘Budrovka' are increasingly explored as flexible upstream systems for plant molecular farming due to their well-characterized monoterpenoid and phenolic pathways and their reliable response to in vitro manipulation. Yet, long-term quantitative descriptions of callus biomass development remain limited. We monitored callus formation from root and stem explants in 3-week intervals across a 15-week cultivation period to generate a structured dataset suitable for predictive modelling. Five algorithms (Beta Regression, GAM, Random Forest, SVR, XGBoost) were trained on mean response data to estimate callus weight and induction efficiency. XGBoost achieved the highest accuracy for biomass prediction ($R2 = 0.9373$; RMSE 1.76 g), and feature-importance profiles consistently identified explant origin, culture duration and auxin-cytokinin regime as dominant contributors to callus performance. NSGA-II multi-objective optimization, guided by GAM and XGBoost outputs, indicated that 15-week root-derived calli grown under 0.5 mg/L 2,4-D and 2 mg/L BAP represent the highest-yielding configuration. The study provides a data-resolved view of long-term callus behavior in hybrid lavender and a modelling-based approach for rational tuning of biomass production in plant-based bioprocesses.; This work was supported by project FPPV-31-2026.

P-29 Wed 13 & Thu 14 May

Medium optimization of Datura stramonium L. root cultures to enhance tropane alkaloid production

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Topic: Scaling Up Plant-Based Production

Plant-based production systems require robust and predictable in vitro platforms to support process optimization and future scale-up. Primary, non-transformed root cultures of Datura stramonium L. were used as a controlled in vitro platform to optimize the biosynthesis of the pharmaceutically relevant tropane alkaloids hyoscyamine and scopolamine. The combined effects of indole-3-acetic acid (IAA) and sucrose concentrations in liquid Gamborg B5 medium were evaluated using DoE and RSM. Biomass accumulation remained statistically comparable across all tested conditions, indicating stable growth kinetics suitable for process optimization. In contrast, alkaloid levels were significantly influenced by the interaction between auxin availability and carbon source concentration. RSM modelling enabled numerical optimization and identified 4 mg/L IAA and 50 g/L sucrose as conditions predicted to enhance alkaloid production while maintaining biomass stability. Experimental validation confirmed model accuracy for scopolamine and growth parameters, whereas hyoscyamine showed higher biological variability, suggesting the need for broader design space in future work. These findings demonstrate that targeted medium optimization can increase metabolite yield independently of biomass changes and support the use of DoE-RSM strategies to improve process predictability in root culture plant production systems. This work was supported by project FPPV-05-2026.

P-30 Wed 13 & Thu 14 May

Successful Production of Human Type II Collagen in Plants and Establishment of a Facility for Scale-Up and R&D Support

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Topic: Scaling Up Plant-Based Production

Biomanufacturing represents a sustainable alternative to conventional chemical processes due to its superior energy efficiency and reduced dependence on fossil resources. Despite this promise, challenges such as limited productivity and constraints in achievable material properties remain. While microbial platforms are often inadequate for producing large, structurally complex proteins, plants offer distinct advantages as hosts for high molecular weight recombinant protein synthesis. In this study, we demonstrate the successful production of human type II collagen--an essential structural component of cartilage--using a plant-based expression system. Because this protein requires intricate post translational modifications, it is difficult to express in standard microbial systems such as E. coli. Our analyses confirmed that the plant-derived collagen retains its native triple helical conformation and exhibits physicochemical properties comparable to those of collagen derived from animal tissues. To promote the commercialization of plant-made biomaterials, we have established a dedicated research and development facility designed to support feasibility evaluation and process scale-up from laboratory to industrial levels. Through this infrastructure, we aim to accelerate the development of diverse high-value plant-derived materials and contribute to the advancement of the plant molecular farming industry.

P-31 Wed 13 & Thu 14 May

Accelerated Breeding of Non-Flowering Tobacco Lines for Biomass Enhancement

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Topic: Smart Chassis - Engineering the Plant Factory

Nicotiana tabacum is a promising plant biofactory due to its high biomass yield, well characterized secondary metabolism, and metabolic compartmentalization. Enhancing biomass is therefore a key goal for engineering tobacco as a chassis for leaf based secondary metabolite production. Nevertheless, obtaining transgene free CRISPR mutants optimized for vegetative growth in tobacco is time consuming due to its relatively long-life cycle. We employed CRISPR Cas mutagenesis of endogenous flowering activator genes to extend vegetative growth and increase biomass. The thermotolerant Cas12 endonuclease generated quadruple flowering activator mutants in a single generation. To accelerate the breeding cycle, a florigen gene was overexpressed within the T DNA, reducing flowering time by half compared to standard protocols and enabling rapid selection of T DNA free CRISPR mutant seeds. Viral vectors overexpressing a florigen gene were also evaluated as a graft free strategy to induce flowering for propagation of non flowering lines. Flowering activator mutants accumulated greater biomass by the time wild type plants initiated flowering and remained in a prolonged vegetative state, unable to flower under long day or short day conditions, with potential for even greater biomass due to continued vegetative growth. This approach provides a versatile strategy to shorten breeding timelines and generate high biomass, non flowering tobacco lines optimized as sustainable plant biofactories.

P-32 Wed 13 & Thu 14 May

Towards Using Synthetic Biomolecular Condensates to Improve IgG Yield in Planta

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Topic: Smart Chassis - Engineering the Plant Factory

Nicotiana benthamiana offers a robust and flexible platform for the production of therapeutic antibodies, including IgGs against Ebola and HIV. Plant-derived IgG yield is, however, limited by protease degradation and further complicated by inefficient secretion of IgG into the apoplast. We hypothesised that targeting IgG to synthetic biomolecular condensates in the secretory pathway will improve IgG yield by: reducing ER stress associated with IgG expression (and thereby enhancing IgG folding), shielding IgG from proteolytic degradation, and improving IgG secretion into the apoplast. Our preliminary data demonstrate that a transiently expressed condensate scaffold recruits its client in N. benthamiana, leading to a change in the yield of the condensate-targeted client. This is a ground-breaking system, showing that synthetic condensates can be successfully targeted to the plant secretory pathway. In the future we will apply this technology to molecular farming, to improve IgG yield by targeting antibodies to synthetic condensates in the plant secretory pathway. This technology has the potential to substantially improve industrial plant molecular farming.

P-33 Wed 13 & Thu 14 May

Metabolic and developmental engineering of Nicotiana tabacum for the enhanced production of terpenoids in glandular trichomes.

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Topic: Smart Chassis - Engineering the Plant Factory

Nicotiana tabacum glandular trichomes produce large quantities of diterpenes, representing a promising and sustainable alternative to conventional systems for producing high-value industrial or pharmaceutical terpenoids. While traditional breeding has generated tobacco varieties with favourable agronomic traits, breeding for biofactory-related traits remains largely unexplored. Furthermore, genetic tools for trichome-targeted metabolic engineering and heterologous compound production are not yet fully optimized. We evaluated trichome density in eighteen high-diterpene-yielding N. tabacum varieties using an automated quantification pipeline to identify a suitable biofactory chassis. We characterized the effects of CRISPR-Cas12 mutations in genes affecting trichome density and diterpene biosynthesis. Finally, we generated dual-reporter transgenic lines to assess the expression of five promoters described as trichome-specific. Trichome density and productivity were uncoupled, contributing independently to diterpene yield. CRISPR-Cas12 mutant lines demonstrated that targeted breeding can increase trichome density and block the main diterpene biosynthetic pathway. Dual-reporter lines revealed unexpected basal promoter expression in leaf tissue, challenging their reported specificity. This work contributes to the development of plant biofactories by integrating variety selection, promoter characterization, and genome editing to enhance terpene yield.

P-34 Wed 13 & Thu 14 May

From plant to bioreactor: a robust workflow for establishing high-performance plant cell cultures

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Topic: Smart Chassis - Engineering the Plant Factory

Traditional production of high-value plant secondary metabolites is frequently hampered by environmental instability and resource-intensive farming. By transitioning to a bioreactor-based production system, we can offer a sterile and reproducible alternative to field cultivation. To move efficiently to a cell culture process we present a streamlined, two-stage strategy for the development of stable plant suspension cell lines. First, callus induction is screened for optimal growth and consistency for suspension cell lines. Second, a systematic macronutrient-based media optimization was implemented for the simultaneous selection of high growth and cell morphology. Using this screen, we successfully established a novel Moringa oleifera-derived cell line specifically engineered for industrial application. The M. oleifera cell line exhibits a low-aggregation phenotype and fast growth kinetics. Beyond its native capacity for bioactive secondary metabolite synthesis, the cell line's high endogenous protein content provides a robust foundation for recombinant protein expression. This research demonstrates how precise medium optimization and cell line selection can decouple plant-derived production from agricultural constraints. The developed M. oleifera platform serves as a versatile, robust biotechnological tool for the pharmaceutical industry.

P-35 Wed 13 & Thu 14 May

Elicitor-Mediated Modulation of Tropane Alkaloid Production in Root-Derived Callus Cultures of Datura stramonium L.

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Topic: Smart Chassis - Engineering the Plant Factory

Callus cultures of Datura stramonium L. offer a controlled in vitro platform for the production of pharmaceutically relevant tropane alkaloids. In this study, root-derived calli were established on Murashige and Skoog medium supplemented with selected auxin-cytokinin combinations and assessed for their ability to accumulate hyoscyamine and scopolamine. Basal calli produced only low alkaloid levels, but elicitation markedly enhanced biosynthesis. A broad range of biotic and abiotic elicitors was tested, with yeast extract (300 mg/L) and a plant extract containing cucumber mosaic virus (CMV) identified as the most effective treatments. Yeast extract increased hyoscyamine accumulation up to 24.7 mg/g DW (approximately 39-fold compared to control), while both yeast extract and the CMV-containing extract enhanced scopolamine levels more than 30-fold. Biomass responses to elicitation varied depending on the plant growth regulator composition used during callus maintenance, with methyl jasmonate (100 mg/L) supporting the largest biomass increase. These results demonstrate that D. stramonium callus cultures can be efficiently stimulated to produce high levels of tropane alkaloids and highlight the importance of selecting suitable elicitors and growth regulator combinations for optimizing metabolite output in plant in vitro production systems. This work was supported by project FPPV-05-2026.

P-36 Wed 13 & Thu 14 May

Next-Generation Genome Editing Tools for Plant Biofactories

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Topic: Smart Chassis - Engineering the Plant Factory

Precision and versatility in genome editing are essential for next-generation plant biofactories. We present two complementary strategies to overcome plant-specific limitations of current CRISPR tools. First, we develop AlCas12a, a novel metagenomic Cas12a nuclease, optimized for plant expression and paired with a TYLCV-based replicon to increase intracellular accumulation. We then apply structure-guided in silico protein engineering in two directions: (i) conserved mutations to obtain thermotolerant AlCas12a variants capable of efficient DNA cleavage at 22-25 °C, and (ii) chimeric and WED-III-corrected variants generated by domain swapping and targeted mutagenesis to restore pre-crRNA processing and enable multiplex genome editing. All variants have been computationally modeled and are being prepared for experimental validation. Second, we evaluate Prime Editing in Nicotiana benthamiana using a macroscopically visible fungal-bioluminescence reporter to directly score precise edits. Prime Editing version 7 was assessed, and editing efficiency was further enhanced using two amplification strategies (i) a TYLCV replicon to enhance editor and guide expression and (ii) a PhiC31 “SWITCH-ON” TMV replicon that amplifies reporter signal only upon successful PhiC31 editing. Together, AlCas12a engineering and system-level Prime Editing optimization expand the CRISPR toolbox for precise, efficient and multiplex genome editing in dicot plants.

P-37 Wed 13 & Thu 14 May

Development of Autobioluminescent Sentinel Plants for the Early Remote Detection of Viral Infections

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Topic: Smart Chassis - Engineering the Plant Factory

Using plants as high-tech biofactories for high-value compound production demands extreme biosecurity and control. In these platforms, a viral infection is not merely a pathology but a critical failure in the production chain. To ensure batch integrity without skyrocketing costs, developing monitoring early-warning systems independent of complex instrumentation is imperative.; This project implements the "sentinel plant" concept in Nicotiana benthamiana for continuous monitoring using simple imaging systems. By coupling the fungal bioluminescence pathway to geminivirus vectors, a sensor with an OFF■ON logic was generated. This configuration detects infection before pathological phenotypes emerge, granting a larger action window. To provide a dedicated analytical tool, LumTrack was developed, a custom software that automates image analysis and visualizes spatiotemporal dynamics.; To expand the spectral detection range, the bacterial bioluminescence pathway coupled with BRET architectures was explored. This study assessed the feasibility of this pathway in plants and its potential to engineer a versatile palette of multispectral reporters.; Overall, this work validates sentinel plants as robust, self-reporting biosensors. By serving as co-cultivated biological proxies, they enable early detection and continuous monitoring, proposing a potentially scalable strategy to secure high-value production crops.

P-38 Wed 13 & Thu 14 May

Bio-encapsulation of immunotherapeutics for allergy treatmentfor allergy treatment

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Topic: The Molecular Farming Marketplace

Allergen-specific immunotherapy is a disease-modifying treatment for respiratory and insect venom allergies, but its application to food allergies remains limited. Conventional subcutaneous immunotherapy is associated with adverse reactions, prolonged treatment periods, and only transient clinical benefits. In contrast, oral immunotherapy induces desensitization and long-term immune tolerance through controlled allergen exposure and has shown promising clinical and immunological outcomes. Safety can be further improved by generating hypoallergenic variants that retain immunological activity while reducing allergenic potential. However, orally administered allergens undergo extensive proteolytic degradation in the gastrointestinal tract. Hypoallergenic parvalbumin, a promising candidate for fish allergy treatment, is particularly unstable upon mucosal delivery and therefore fails to induce durable oral tolerance. To overcome this limitation, we bioencapsulated hypoallergenic parvalbumin within zein protein bodies using Nicotiana benthamiana as an expression host. Zein protein bodies are naturally occurring seed storage organelles with high mechanical and chemical stability. A multilayered protein body platform composed of three distinct zeins enabled spatial targeting of the allergen to defined protein body layers. Protein bodies were produced at larger scale and evaluated in a murine food allergy model, where oral administration resulted in encouraging immunological outcomes

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P-39 Wed 13 & Thu 14 May

Plant-made basic Fibroblast growth factor (bFGF) for the usage in animal cell culture

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Topic: The Molecular Farming Marketplace

The anticipated rise in global meat consumption, combined with increasing pressure to reduce greenhouse gas emissions and improve animal welfare, has intensified work in cultivated meat as a sustainable protein source. A major bottleneck remains the cost-effective, animal-free cell culture media. Fetal bovine serum, the source of growth factors and carrier proteins, is incompatible with the ethical, regulatory, and scalability requirements of cultivated meat production. Molecular farming offers a scalable alternative for producing recombinant growth factors. Basic fibroblast growth factor (bFGF) is a key cytokine regulating cell proliferation and lineage commitment and represents a major cost driver in serum-free media. In this study, porcine bFGF variants were produced via transient expression in N. benthamiana using the viral MagnICON expression system. Different gene constructs were evaluated with respect to expression yield, purification efficiency, and biological activity to identify optimal production strategies. The most promising bFGF variants were transferred into pea (Pisum sativum) for stable, seed-specific expression. Eleven independent transgenic pea events were generated, and bFGF accumulation was quantified in homozygous lines. This work establishes peas as a platform for seed-based production of functional animal growth factors. Future studies will investigate whether minimally processed seed extracts can be directly applied in animal-free cell culture media.

P-40 Wed 13 & Thu 14 May

Plant-made viral nanoparticles for molecular farming: Insights into RNA inactivation strategies and RNA-free alternatives

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Topic: Viral Vectors and Nanoparticles in Molecular Farming

Plant-made nanoparticles represent a versatile therapeutic strategy in the pharmaceutical field. In the context of autoimmunity, Tomato Bushy Stunt Virus (TBSV) was engineered to display a pLiprin (pLip) peptide, reported as auto-dominant for rheumatoid arthritis, produced in Nicotiana benthamiana, and its functionality explored as platform for the induction of immune tolerance. This single-stranded RNA-containing virion, while suitable for therapeutic reasons, may represents a bioburden that would require specific and costly disposal, especially when produced in large scale for commercial purposes. Therefore, RNA-inactivation strategies on the final product were investigated, in particular, the use of ultraviolect light (UV) and Binary Ethylenemine (BEI). UV inactivation is a commonly employed technique to damage nucleic acids, while BEI is useful to prevent viral replication, while maintaining the protein structure of the capsid. These strategies would both maintain virus antigenicity, while inactivating its RNA, resulting in a safe and environmentally-friendly active pharmaceutical ingredient (API). On this note, the production of empty virus-like particles (eVLPs) was attempted, to directly obtain an RNA-free virion, to bypass viral inactivation steps. These strategies could innovate and reduce the environmental risk of the API, while maintaining its therapeutic potential, following appropriate validation.

P-41 Wed 13 & Thu 14 May

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Topic: Viral Vectors and Nanoparticles in Molecular Farming

Exploring a novel CSDaV-derived VLP platform for biotechnological applications and production in microalgae Virus-like particles (VLPs), including those derived from plant viruses, can be used in biopharmaceutics as a scaffold for vaccines or as carriers by cargo encapsulation. In this work, we introduce the opportunities and limitations of using citrus sudden death-associated virus (CSDaV)-derived VLPs as a novel biotechnological platform. CSDaV-VLPs can be produced in N. benthamiana plants by expressing CSDaV's major coat protein (CP), with particle sizes and morphology similar to authentic virions. Furthermore, attempts to fuse small peptide sequences to the CSDaV-CP demonstrated that CSDaV-CP only allows for N-terminal genetic fusions to maintain VLP formation. By using the SpyTag/SpyCatcher system, GFP can be bound to VLPs, where it is presented in the internal cavity of the particle. Currently, conditions for dis- and re-assembly of CSDaV-VLPs are being investigated for cargo encapsulation. In addition, a scalable microalgae-based production platform is being explored to enable the production of plant VLPs. To this end, CP genes derived from two plant viruses, CSDaV and CCMV, have been cloned into a microalgae expression vector and are currently being transformed in microalgae. The outcome of this project aims at a proof-of-concept platform combining plant virus-derived VLPs and a microalgae production system to enable fast and customizable biopharmaceutical engineering

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P-42 Wed 13 & Thu 14 May

It's All about Networking: Engineered Plant Virus Nanoparticles for cross-linking of hydrogels

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Topic: Viral Vectors and Nanoparticles in Molecular Farming

In tissue engineering, hydrogels mimic the extracellular matrix, that interacts with and modulates cell behavior. Current research focuses on developing hydrogels that provide dynamic biological and mechanical cues, modulated by the embedded cells during tissue maturation. Plant virus nanoparticles (VNPs) have emerged as promising nanocomposite candidates for hydrogels due to their ability to present diverse peptides on their surface, enhancing cell support and hydrogel properties. Here, genetically engineered and plant-produced potato virus X (PVX) nanoparticles serve as co-building blocks in a gelatin-based hydrogel system. These VNPs are designed with motifs facilitating crosslinking via microbial transglutaminase (mTG). The integrity of the VNPs was confirmed through Western blotting, RT-PCR and transmission electron microscopy. Upon incubation with mTG and gelatin, crosslinking was validated using SDS-PAGE and Western blot analysis. The proteo- and hydrolytic stability alongside mechanical properties was characterized, revealing enhanced stability at a 1/600 VNP-to-gelatin ratio. Live/dead staining indicated maintained cell viability, while immunofluorescent labeling demonstrated increased cell interactions.These findings highlight the potential of engineered PVX as biocompatible nanocomposites that stabilize hydrogels and create a dynamic microenvironments mimicking natural extracellular matrices.

P-43 Wed 13 & Thu 14 May

Plant virus nanoparticles as biocompatible delivery system for siRNA to selectively control pests

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Topic: Viral Vectors and Nanoparticles in Molecular Farming

The increasing demand for sustainable agricultural practices has highlighted the need for innovative solutions in plant protection. RNA interference has been utilized to develop RNA-based pesticides as an alternative to traditional chemical formulations. However, due to the low environmental RNA stability, effective nucleic acid delivery requires safe, stable, and efficient vehicles to protect these molecules from degradation. Synthetic nanoparticles involve high manufacturing costs as well as safety and environmental concerns. In contrast, engineered plant virus-like particles (VLPs) can be used to encapsulate small interfering RNA (siRNA) to create efficient and biodegradable biopesticides that target plant pathogens without the need for transgenic plants. Here, we demonstrate the successful in planta and in vitro assembly of the potato virus X coat protein and non-viral RNA targeting GFP. The structures of different siRNA/VLPs are assessed using transmission electron microscopy. The localization and silencing capacity of the sprayed siRNA/VLPs in GFP-transgenic plants are determined by confocal microscopy, RT-PCR and Western blotting. In future, RNA size and concentration in filamentous and spherical VLPs will be optimized to induce potent silencing effects. This work highlights siRNA/VLPs as biocompatible, non-infective pest management system convenient for farmers.

P-44 Wed 13 & Thu 14 May

Improving the tobacco mosaic virus vector for systemic expression of recombinant proteins in plants

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Topic: Viral Vectors and Nanoparticles in Molecular Farming

Viral vectors have been established as one of the most rapid and efficient plant-based systems for recombinant protein production, with tobacco mosaic virus (TMV) notably achieving remarkable accumulation levels of some proteins of interest. Despite this potential, TMV-based platforms are traditionally limited by a narrow cargo capacity and a lack of systemic mobility; since the protein of interest typically replaces the viral coat protein (CP), the resulting vector is unable to move beyond the initial infection site. To address these bottlenecks, this work explores strategies to decouple protein expression from movement constraints by leveraging the correlation between expression levels and movement protein (MP) availability. We generated two transgenic Nicotiana benthamiana lines designed to provide viral functions in trans: the first expresses the full-length MP to decompress the viral genome and allow for larger exogenous inserts, while the second expresses a gain-of-function MP mutant, lacking the 16 C-terminal amino acids, facilitating systemic viral spread even in the absence of the CP. Complementing these host-based strategies, we developed a novel vector utilizing the P2A self-processing peptide to fuse the recombinant protein to the CP. This approach preserves CP integrity and ensures systemic movement in wild-type plants. Together, these systems provide a flexible toolkit for high-yield, systemic production of large and complex proteins in plant-based biofactories.

P-45 Wed 13 & Thu 14 May

Viral vectors for plant metabolic engineering

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Topic: Viral Vectors and Nanoparticles in Molecular Farming

Plant viruses are frequently reshaped into expression vectors for the efficient and quick production of recombinant proteins in plants, which are used as bioreactors. Interestingly, plant viruses can also be engineered to produce metabolites of interest in biofactory plants. To do so, virus-based vectors typically express transcription factors or metabolic enzymes that impact into endogenous or novel metabolic pathways in the host plants. Our research group has pioneered the use of viruses within the genus Potyvirus, such as tobacco etch virus, to produce anthocyanins, carotenoids and apocarotenoids in Nicotiana benthamiana. The use of potyviral vectors in plant metabolic engineering is based on the relatively large cargo capacity of this type of elongated, flexuous RNA viruses, and the efficient and highly specific activity or the viral nuclear inclusion a protease (NIaPro). Our current research aims to open the technology to other potyviruses, such as lettuce mosaic virus or beet mosaic virus, which may allow to produce the metabolites of interest in edible plants, and to explore the use of minus-strand RNA viruses, such as tomato spotted wilt virus. Based on particularities of the infection cycle, this kind of vectors may exhibit larger cargo capacity with higher genetic stability.;

P-46 Wed 13 & Thu 14 May

Coupling of Replication and Selective Packaging in Cowpea Mosaic Virus Enables Custom RNA Encapsidation for Biopesticides

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Topic: Viral Vectors and Nanoparticles in Molecular Farming

Cowpea mosaic virus (CPMV) is a bipartite plant RNA virus that provides a unique opportunity to integrate fundamental virology with plant molecular farming. In CPMV, selective encapsidation is intrinsically linked to replication, such that only replication-competent RNAs are packaged into viral particles. Exploiting this, heterologous RNAs flanked by CPMV untranslated regions can be efficiently replicated and selectively packaged into virus-like particles (VLPs) that are readily purified from agroinfiltrated plant tissue. We show here that a long hairpin RNA (forming a pseudo double-stranded RNA) encoding inverted repeats of a gene sequence from an agricultural insect pest (Colorado Potato Beetle) is efficiently and stably encapsidated within CPMV VLPs, with integrity retained after purification, underscoring their potential as dsRNA delivery vehicles. Notably, packaging is significantly more efficient when a version of RNA-1 is used which cannot replicate itself, but is only capable of replication in trans. These findings validate CPMV VLPs as a potential platform for Spray-Induced Gene Silencing (SIGS), a highly specific and environmentally sustainable pest-control strategy based on RNA interference and gene silencing following dsRNA ingestion by target pests. Collectively, this work positions CPMV dsRNA VLPs as a next-generation RNA delivery technology which we intend to combine with industry-enabled formulation pipelines for pest control product development.

P-47 Wed 13 & Thu 14 May

Characterization of Nicotiana-derived signal peptides and promoters for molecular farming

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Topic: The Molecular Farming Marketplace

Plant-based expression systems are widely used in molecular farming, but their efficiency strongly depends on the regulatory and targeting elements employed. This study aimed to characterize a panel of signal peptides and promoters from Nicotiana benthamiana and Nicotiana tabacum for recombinant protein production. Several endogenous plant proteins that accumulated in the apoplast in response to agroinfiltration were identified, and their associated signal peptides and promoters were selected for evaluation. Their performance was tested in transient expression assays using different recombinant proteins and compared with commonly used signal peptides and promoters in plant biotechnology. Our results suggest substantial variability in performance, particularly for signal peptides, which appears to be protein-dependent. One promoter showed especially promising results across constructs. These findings contribute to expanding the molecular toolbox available for plant molecular farming and support the use of endogenous Nicotiana regulatory elements as valuable alternatives for recombinant protein expression.

P-48 Wed 13 & Thu 14 May

CRISPR Editing of a Light-Responsive Pathway Unlocks Polyphenol Biosynthesis in Annurca Apple

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Understanding how environmental signals modulate secondary metabolism remains a key challenge in plant biology. In this study, we combined CRISPR-Cas9 gene editing with callus culture systems to investigate the genetic and environmental regulation of the dihydrochalcone and flavonoid pathways in Malus domestica cv. Annurca. This traditional apple cultivar from Southern Italy is renowned for its rich polyphenolic profile, making it an ideal model to study light-dependent metabolic regulation. Peel-derived callus cultures provided a rapid and controllable platform for targeted gene manipulation and metabolic analysis. Following MdCHI gene knockout, both control and edited lines were grown under light and dark conditions to examine how light influences pathway activation. Expression profiling of key biosynthetic genes (Md4CL, MdCHS, MdCHI, MdUGT, and MdFLS) showed that light strongly induces their expression, causing pronounced differences in polyphenol and flavonoid accumulation, particularly in edited lines. These results demonstrate that combining gene editing and in vitro culture systems enables functional dissection of environmentally responsive biosynthetic networks and enhances our understanding of light-regulated secondary metabolism. Future RNA-seq analyses will identify transcription factors regulated by light and involved in activating polyphenol pathway genes in Annurca apple, providing new insights for targeted manipulation of secondary metabolism in fruit crops.

P-49 Wed 13 & Thu 14 May

Controlling heterologous protein synthesis through a plant RNA ThermoSwitch

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Topic: Smart Chassis - Engineering the Plant Factory

Among the most exciting recent discoveries are plant RNA ThermoSwitches, embedded within 5′ untranslated regions (UTR) of several mRNAs. These cis-acting RNA elements sense temperature shifts and instantly tune translational output. First uncovered in Arabidopsis, ThermoSwitches are emerging as a powerful new frontier for biotechnology, offering a programmable way to modulate protein production through temperature adjustments. Using a dual fluorescence reporter construct introduced by Agrobacterium-mediated transient expression, we show that the ThermoSwitch from the 5′ UTR of the Arabidopsis PIF7 mRNA functions as a temperature-responsive module in N. benthamiana leaves. A shift from 17 to 27 °C increased reporter translation, delivering ~70% higher output within one day. This magnitude mirrors the rise in endogenous PIF7 translation observed in Arabidopsis following the same shift. The response remained even when the upstream 5′ UTR context was replaced with an unrelated sequence, showing that the ThermoSwitch operates autonomously. This work provides the first evidence that a plant RNA ThermoSwitch functions effectively in a transient expression system, establishing a homogeneous, temperature-responsive gene regulation system free from chemical inducers or repressors. The autonomous operation of the Arabidopsis ThermoSwitch in N. benthamiana highlights its value as a versatile module that can be deployed in plant systems for broad biotechnological applications.

P-50 Wed 13 & Thu 14 May

Targeted knockout of barley endosperm-specific storage proteins as a prerequisite for molecular farming purposes

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Topic: Smart Chassis - Engineering the Plant Factory

Molecular farming enables the production of high-value recombinant proteins in plants. While transient expression systems in tobacco are widely used, cereal grains offer an attractive alternative as stable bioreactors capable of storing proteins under ambient conditions. The starchy endosperm provides a scalable and cost-effective platform for recombinant protein accumulation. To increase recombinant protein yield and reduce competition from endogenous storage proteins, CRISPR/Cas9-mediated knockouts of barley Hordein B1 family members were produced. Mutant plants were validated using Sanger and deep amplicon sequencing, and grain morphology was assessed through automated phenotyping. As a proof of concept, recombinant human Epidermal Growth Factor (EGF) was expressed in wild-type and horb1 mutant backgrounds. The horb1 grains showed altered biometric traits, reduced total protein and hordein content, and delayed germination compared to wild-type segregants. Western blot and ELISA analyses demonstrated increased EGF accumulation in the mutant background. These findings indicate that reducing endogenous storage proteins can enhance recombinant protein accumulation in barley grains, though further optimisation will be necessary to minimise germination penalties.

P-51 Wed 13 & Thu 14 May

Biomanufacturing Synergies of Brassica oleracea and Cinnamomum verum for Targeted Steroidogenic Modulation

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Topic: Scaling Up Plant-Based Production

Currently, Plant Molecular Farming (PMF) enters an era of precision bio-manufacturing, plants are increasingly utilized as bioreactors for complex-trait therapeutics. This study investigates the phytochemical factories of Brassica oleracea and Cinnamomum verum in modulating mammalian steroidogenesis, specifically serum Estradiol (E2) and Progesterone (P4). In a four-week longitudinal model, we analyzed these bioactives in isolation and synergy. Results show the cinnamaldehyde and cinnamic acid profile of C. verum probably acts as a high-potency P4 modulator; in females, P4 surged from 0.12 ng/ml to 0.46 ng/ml by Week 2. Simultaneously, a "Hormonal Protective Effect" was identified in the combination group. We hypothesize that sulforaphane and indole-3-carbinol from B. oleracea stabilize estrogenic pathways; while controls saw E2 decline to 5.26 pg/ml, the synergistic formulation stabilized levels at 6.86 pg/ml. Notably, males showed intense sensitivity, with P4 reaching 0.69 ng/ml in Week 1. These findings underscore the capacity of plant secondary metabolites to restore hormonal homeostasis. This research provides a proof-of-concept for PMF's future: moving beyond single-protein production toward engineered, synergistic bioactive cocktails as bio-manufactured alternatives to synthetic therapies.

P-52 Wed 13 & Thu 14 May

Quail immunological response to Porcine Circovirus

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Current knowledge on the immunological response of Japanese Quails during viral infection is limited. The study focused on understanding the biology of Porcine Circovirus type 2 (PCV-2) by evaluating the evolution and adaptive immune response in quails using single-cell RNA sequencing and computational methods for data mining. Porcine Circovirus type 2 has had devastating effects on the swine industry, with a direct impact on the socio-economic conditions of many countries. PCV Virus-like particles (VLPs) were assessed by transmission electron microscopy and immunized into a group of quails. A specific antibody response was elicited and detected in both the blood and egg samples. 5 000 purified transcriptome and PBMCs from the immunized quails were separated into individual droplets, RNA extracted, and cDNA synthesised using the Genomics 10X Chromium platform. Analyses revealed 14 clusters, annotated using specific marker genes and differentially expressed genes (DEGs). Cells of both the innate and adaptive immune system was present. High CD4+ T cell and macrophage counts, with inflammatory responses, indicate the development of PCV-2. CD4+ T cells differentiated into Th1-like and Th2-like cells.

P-53 Wed 13 & Thu 14 May

Dissecting gene regulatory networks of plant specialized metabolism and trichome development in Cannabis sativa: exploring the potential effects of hormonal elicitation

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Topic: Metabolic Pathway Engineering for High-Value Compounds

Cannabis sativa has long been cultivated for recreational use, traditional medicine, and fiber production. Today, it is increasingly studied for its therapeutic potential as the primary source of cannabinoids, which have psychoactive and pharmacological effects in human diseases. Cannabinoids mainly accumulate in glandular trichomes (GT) of mature female flowers, alongside with mono- and sesqui-terpenoids. However, the gene regulatory networks controlling cannabinoid/terpenoid biosynthesis and GT development remain poorly understood. We identified candidate transcription factors (TFs) by mining public transcriptomic datasets and used DNA Affinity Purification sequencing (DAP-seq) to examine their genome-wide binding profiles. We then integrated these results with public transcriptomic datasets to predict the target genes of TFs. Since phytohormones such as jasmonate and gibberellins are known to regulate trichome development and terpene metabolism in other plant species, we also conducted hormonal elicitation experiments in whole plants and cell suspension cultures from cannabis genotypes with diverse cannabinoid profiles. Our findings suggest that R2R3-MYB TFs, specifically CsMYB24 and CsMIXTA, may regulate terpene accumulation and GT differentiation in female cannabis flowers. Our work aims provides new insights for enhancing metabolite production in whole plants and plant cell cultures, supporting future advances in the biotechnological applications of cannabis.

P-54 Wed 13 & Thu 14 May

Rewriting nature's chemistry

Barbara Radzikowska

Topic: Metabolic Pathway Engineering for High-Value Compounds

HotHouse Therapeutics is advancing drug discovery with powerful AI-driven plant bioengineering platforms creating complex molecules beyond the reach of conventional chemical synthesis approaches. By engineering biosynthetic pathways in plants, we generate novel, high-purity compounds, including next-generation vaccine adjuvants, rapidly and sustainably. Our integrated platforms, BotanAI and BotanBio, deliver novel molecules at scale, including promising new adjuvants, through a sustainable and cost-effective supply chain. From pioneering vaccine adjuvants to entirely new classes of therapeutics, HotHouse Therapeutics is building not just a pipeline, but a platform to redefine what's possible in drug discovery, driving both scientific innovation and commercial impact.

P-55 Wed 13 & Thu 14 May

Integrating AI-Driven Protein Design into Plant-Based Expression Platforms for Next-Generation Advanced Therapeutics

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Plant expression systems enable complex biologic production with tailored glycosylation and scalability. We previously demonstrated plant-derived EpCAM fused to IgG, IgA, and IgM Fc domains as effective vaccine candidates. Through Agrobacterium-mediated transformation of Nicotiana tabacum and cross-fertilization, we successfully expressed high-order quaternary structures with ER-retention motifs and Joining chains. Notably, the EpCAM-IgM FcK fusion protein significantly activated dendritic cells in vitro and in vivo, inducing robust immune responses. These proteins exhibited superior binding affinity and inhibited cancer cell proliferation in PC-3 and SW620 models. Building on this platform, our research is expanding toward AI-driven vaccine and antibody development for emerging infectious diseases. We are integrating computational modeling and deep learning at the University of Toronto to optimize antibody stability and target-specific neutralizing epitopes. This approach aims to create potent vaccine candidates and therapeutic antibodies with enhanced structural precision for rapid pandemic response. Combining AI-based predictions with plant expression platforms, we are developing a next-generation pipeline for advanced biologics. This presentation highlights our success in plant-made vaccines and the transition toward AI-augmented engineering for global vaccine and antibody preparedness.

P-56 Wed 13 & Thu 14 May

Co-transient expression of anti-HER1/HER3 bispecific antibodies in Nicotiana benthamiana using the Knobs-into-holes strategy

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

HER1 (EGFR) and HER3 are HER receptor family members that form heterodimers to activate PI3K/Akt signaling, promoting tumor cell survival and proliferation. Ligand binding by EGF and HRG enables compensatory crosstalk that contributes to resistance against HER1-targeted therapies. Although HER3 lacks intrinsic kinase activity, it serves as a critical adaptor upon heterodimerization. To overcome this resistance, bispecific antibodies that simultaneously target HER1 and HER3 represent a promising strategy to disrupt ligand-induced dimerization. In this study, we designed a bispecific antibody composed of anti-HER1 scFv-FcK (Knob) and anti-HER3 scFv-FcK (Hole), transiently co-expressed in Nicotiana benthamiana using the Knobs-into-holes (KiH) strategy. Expression and purification were confirmed by Western blot, SDSPAGE, and HPLC analyses. ELISA, FACS, and cell-based assays verified dual binding to HER1 and HER3, while wound healing and MTT assays demonstrated inhibition of cell migration and proliferation. These findings highlight the therapeutic potential of plant-derived bispecific antibodies in targeting HER1/HER3-driven epithelial cancers.

P-57 Wed 13 & Thu 14 May

Modification of human IgG Fc to enhance the half-life of plant-derived antibreast cancer antibody anti-HER2 VHH-FcK

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

HER2-positive breast cancer, characterized by overexpression of the HER2 protein (encoded by the ERBB2 gene), occurs in approximately 15-50% of breast cancer cases. While the current standard treatment, trastuzumab, has significantly improved survival, there are limitations such as high cost, resistance in some patient populations, and reported cardiac toxicity. Recently, research has been actively conducted on antibody-drug conjugates (ADCs), biomarker-based therapeutic strategies, and mechanisms for overcoming resistance. To address these challenges, plant-derived therapeutic protein production is emerging. In this study, we produced an anti-HER2 VHH-FcK from tobacco plants and aimed to enhance its in vivo half-life. Point mutations were introduced into the Fc region of the antibodies. Fc variant proteins were produced and purified from Nicotiana benthamiana using a transient expression system. Anti-cancer/metastatic effects of antibodies were confirmed in vitro through ELISA, cell-based ELISA, FACS, and ADCC assays. These results confirmed the applicability of plant systems to produce anti-cancer treatment candidates against HER2-positive breast cancer.

P-58 Wed 13 & Thu 14 May

Production of hepatitis B virus-like particles in plants for injection-oral immunization

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Virus-like particles (VLPs) mimic native viruses but are not infectious due to lack of genetic material, while retaining ability to induce strong and specific immune response. Last years plants have emerged as versatile expression systems for VLPs production. This enables development of plant-based vaccines against hepatitis B virus (HBV), using VLPs composed of the small surface antigen (S-HBsAg) and/or the core antigen (HBcAg). VLPs can be produced both via transient expression using pEAQ-based vectors or via Agrobacterium-mediated stable nuclear transformation with modified pGPTV vectors. Elasticity of these approaches allows for optimization of production depending on characteristics of a target antigen. This is particularly important for S-HBsAg, a membrane protein which production is much more effective in stable transformation system, although typically remains lower than HBcAg. HBcAg obtained in Nicotiana benthamiana via transient expression and S-HBsAg produced in transgenic Nicotiana tabacum can be purified for injectable vaccines. Despite comparable expression levels the recovery and stability of HBcAg can be up to ten times higher than those of S-HBsAg. Transgenic edible plants, such as lettuce, constitute also oral vaccine platforms. Leaf tissue can be lyophilized without compromising stability of antigens. Flexibility of formulations for both antigens ensures the use for particular applications, including injection-oral therapeutical co-vaccination

P-59 Wed 13 & Thu 14 May

Modification of Fc-fusion protein structures to enhance efficacy of cancer vaccine in plant expression system

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Topic: Plant-Made Biologics - From Antibodies to Advanced Therapeutics

Epithelial cell adhesion molecule (EpCAM) fused to IgG, IgA and IgM Fc domains was expressed to create IgG, IgA and IgM-like structures as anti-cancer vaccines in plant. High-mannose glycan structures were generated by adding a C-terminal endoplasmic reticulum (ER) retention motif (KDEL) to the Fc domain (FcK) to produce EpCAM-Fc and EpCAM-FcK proteins in transgenic plants. Cross-fertilization of EpCAM-Fc (FcK) plants with Joining chain (J-chain, J and JK) plants led to stable expression of large quaternary EpCAM-IgA Fc (EpCAM-A) and IgM-like (EpCAM-M) proteins. Immunoblotting, SDS-PAGE and ELISA demonstrated that proteins with KDEL had higher expression levels and binding activity to anti-EpCAM IgGs. IgM showed the strongest binding among the fusion proteins, followed by IgA and IgG. Sera from the immunized BALB/c mice produced anti-EpCAM IgGs. Flow cytometry indicated that the EpCAM-Fc fusion proteins significantly activated CD8+ cytotoxic T cells, CD4+ helper T and B cells, particularly with EpCAM-FcKP and EpCAM-FcP (FcKP) × JP (JKP). The induced anti-EpCAM IgGs captured human prostate cancer PC-3 and colorectal cancer SW620 cells. The sera inhibited cancer cell proliferation, migration and invasion; down-regulated proliferation markers (PCNA, Ki-67) and epithelial-mesenchymal transition markers (Vimentin); and up-regulated E-cadherin. These findings suggest that plant can produce effective vaccine candidates to induce anti-cancer immune responses.

P-60 Wed 13 & Thu 14 May

Computationally Guided Selection of Carotenoid Cleavage Dioxygenases for Retinal Production in Rice

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Topic: Metabolic Pathway Engineering

In rice, carotenoids and apocarotenoids are important metabolic engineering targets because of their nutritional and functional value. Apocarotenoids are formed through the catalytic activity of carotenoid cleavage dioxygenases (CCDs). However, plant-derived CCDs capable of converting beta-carotene to retinal have not been reported. This study aimed to identify an effective beta-carotene-cleaving enzyme and assess its potential for retinal production in rice. Candidate enzymes from plant (OsCCD4a), animal (HsBCDO1), and bacterial (MbBLH) origins were comparatively examined to identify the most suitable enzyme for retinal production in rice. Structural and functional differences among the enzymes were investigated using a computational pipeline consisting of 3D structure-based phylogenetic analysis, ligand docking, tunnel and pocket analysis, and molecular dynamics simulations. Based on these in silico analyses, beta-carotene cleavage activity of each enzyme was experimentally evaluated using a beta-carotene-producing E. coli system. The selected enzyme was further applied to an integrated rice construct coupling beta-carotene biosynthesis with downstream retinal production. Computational analyses revealed clear differences among the candidate enzymes in protein-ligand interactions and predicted cleavage efficiency. MbBLH was identified as the most efficient enzyme for beta-carotene cleavage based on its structural characteristics. Consistent with these predictions, E. coli expressing MbBLH showed the highest beta-carotene cleavage activity and the greatest amounts of retinoid production among the tested enzymes. Currently, in rice calli carrying the integrated beta-carotene to retinal pathway construct, beta-carotene levels were markedly reduced, while retinal levels increased compared with the beta-carotene-producing control. MbBLH was identified as the most promising enzyme among the candidates tested. These findings provide a basis for engineering retinal production in rice and demonstrate the usefulness of a computationally guided pipeline for enzyme selection in metabolic engineering.

P-61 Wed 13 & Thu 14 May

An Integrated Synbiotic Strategy for the Biocontrol of Citrus and Biopreservation of Citrus Fruits Using Selected Microorganisms and Citrus-Derived Biomolecules

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Topic: Molecular Farming Marketplace

Citrus fruits represent a major agricultural commodity worldwide, yet their production and postharvest handling are significantly affected by microbial diseases and quality deterioration, leading to substantial economic losses. Conventional chemical treatments raise concerns related to environmental impact, human health, and the development of resistant pathogens. In this context, the present study proposes an innovative and integrated approach combining biocontrol and biopreservation strategies based on the synergistic use of beneficial microorganisms and citrus-derived bioactive compounds. Selected bacterial strains, notably Bacillus amyloliquefaciens CB10 and Pseudomonas luteola OB7, previously isolated and characterized for their strong antagonistic activity, were evaluated against major citrus pathogens including Penicillium digitatum, Penicillium italicum, and Geotrichum citri-aurantii. In vitro and in vivo assays on orange fruits demonstrated significant inhibition of fungal growth, confirming the high biocontrol potential of these strains. Additionally, OB7 exhibited notable activity against Phytophthora citrophthora, a key phytopathogen affecting citrus crops. In parallel, bioactive compounds extracted from citrus by-products (peels and seeds), rich in polyphenols, flavonoids, and essential oils, were investigated for their antimicrobial and antioxidant properties. The integration of these biomolecules with the selected microbial strains led to the development of a synbiotic formulation applied as an edible coating on citrus fruits. This combined approach not only enhanced the inhibition of postharvest pathogens but also contributed to the preservation of fruit quality by reducing weight loss, maintaining firmness, and delaying senescence. The proposed strategy highlights the added value of citrus processing by-products and promotes a circular economy model. Overall, this study provides a sustainable and eco-friendly alternative to chemical treatments, integrating preharvest biocontrol and postharvest biopreservation within a unified framework.

P-62 Wed 13 & Thu 14 May

Advancing Plant-Based Biomanufacturing of Recombinant Animal Proteins in Soybean Seeds

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Topic: Scaling Up Plant-Based Production

The increasing demand for high-value animal-derived proteins across food, cosmetic, and biomedical industries has driven the need for scalable and sustainable production platforms. Plant molecular farming offers a versatile alternative to conventional systems, enabling cost-effective production with reduced environmental impact. In this study, we explore soybean as a host for the expression of complex recombinant animal proteins using a seed-based production strategy. Gene constructs were designed and optimized for plant expression and initially evaluated through transient expression in Nicotiana benthamiana to assess protein accumulation, integrity, and expression efficiency. Selected candidates were subsequently introduced into soybean to generate stable transgenic lines, with expression targeted to seeds to exploit their natural capacity for protein storage and long-term stability. Protein expression and accumulation were characterized using biochemical and proteomic approaches, including electrophoretic analysis and mass spectrometry. In addition, strategies to enhance protein stability, post-translational modification, and downstream recovery were investigated. The use of seed-specific expression systems, combined with optimized extraction and purification workflows, supports the development of a robust and scalable production platform. Our findings demonstrate the potential of soybean as an efficient bioreactor for the production of complex recombinant proteins, highlighting its applicability for industrial-scale biomanufacturing. This work contributes to the advancement of plant-based production systems as sustainable alternatives for generating high-value proteins.